Case 1: Events of signal used to train the model (10 events of M1000)
Case 2: Events of signal not used to train the model (10 events of M2500)
Case 3: A mixture two classes of signals, one of which is used to train the model (10 events of SM +10 events of M500)
Case 4: A mixture of two classes of signals, none of which is used to train the model (5 events of SM + 10 events of M750)
Case 5: A mixture of two classes of signals, none of which is used to train the model (5 events of SM + 10 events of M600)

I have sampled n number of events for each of the above case. For eg. M1000_df = M1000_500Ev.sample(n=10)

Note - This pdf contains plots for each case with the following order: Case number, SOM distribution, m1 x m1 score histogram without SM score cut, m1 x m1 Scores (w/o cut and with cut), m1 x m1 with SM score cut, trilep mass plot, m2 x m2 score histogram without SM score cut, m2 x m2 scores (w/o and with SM cut), m2 x m2 scores with SM score cut, trilep mass plot (same as above).

For 40 x 40 SOM model, m1 = 3 and m2 = 5
For 100 x 100 SOM model, m1 = 5, m2 = 11

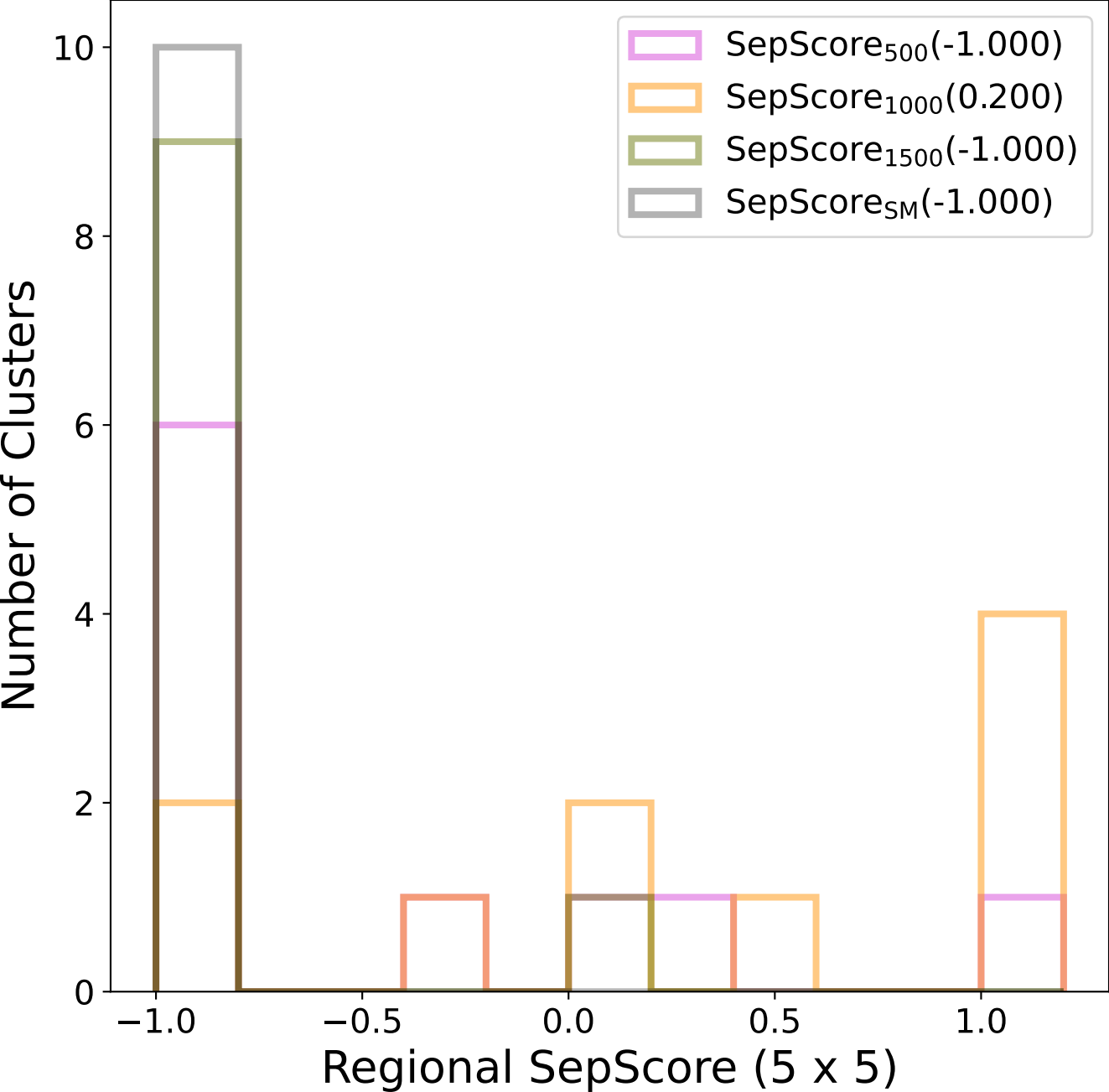
Case 1

Neuron Y

100
95
90
85
80
75
70
65
60
55
50
45
40
35
30
25
20
15
10
5
0

0 5 10 15 20 25 30 35 40 45 50 55 60 65 70 75 80 85 90 95 100

Neuron X


SepScore500(-1.000)
SepScore1000(0.200)
SepScore1500(-1.000)
SepScoreSM(-1.000)
Number of Clusters
0
2
4
6
8
10
−1.0
−0.5
0.0
0.5
1.0
Regional SepScore (5 x 5)

Case 1: cluster of +- 2 around the point

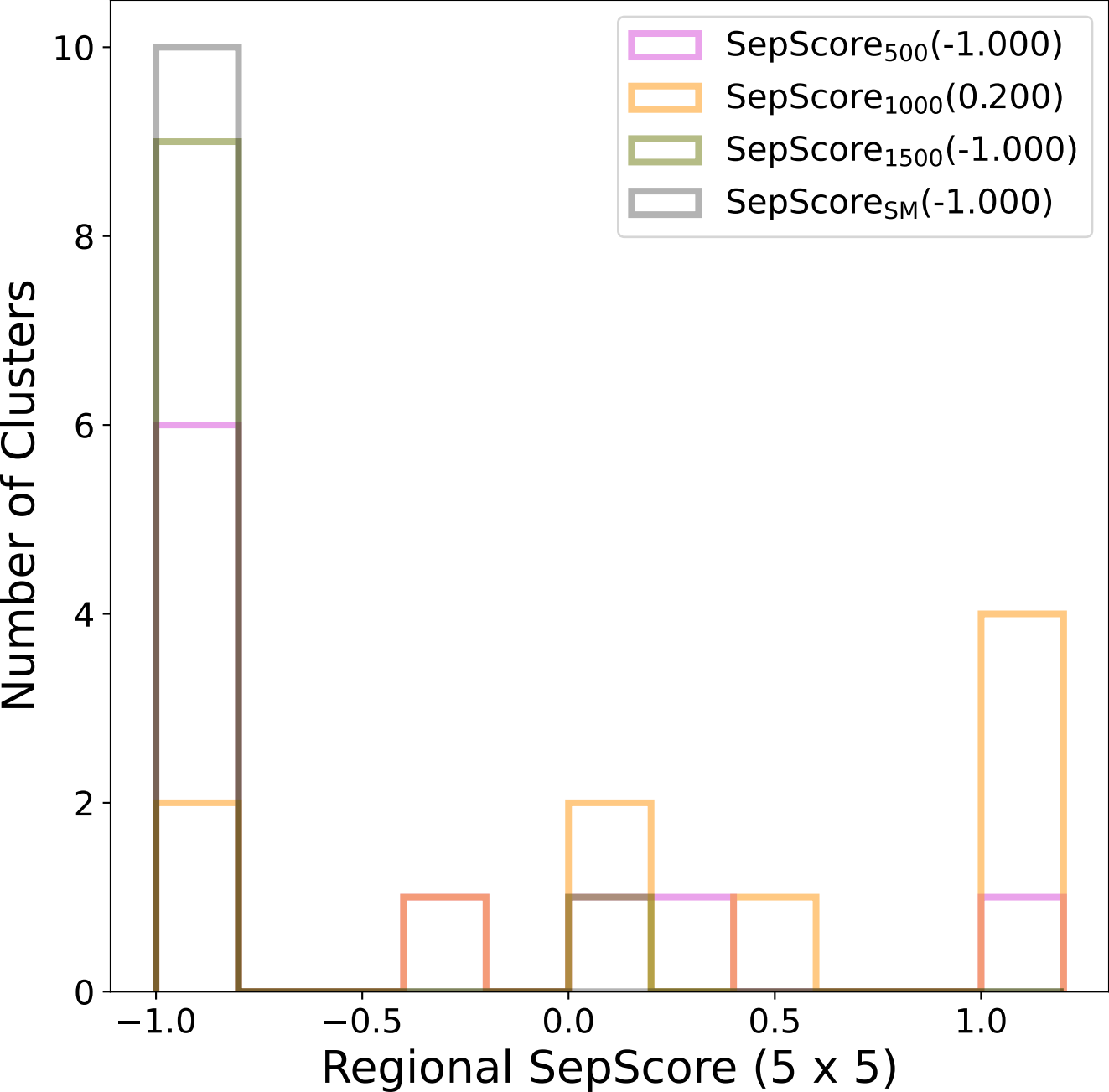
Array of label and scores BEFORE ssSM < 0.6 cut: [label, ss500, ss1000, ss1500, ssSM]:

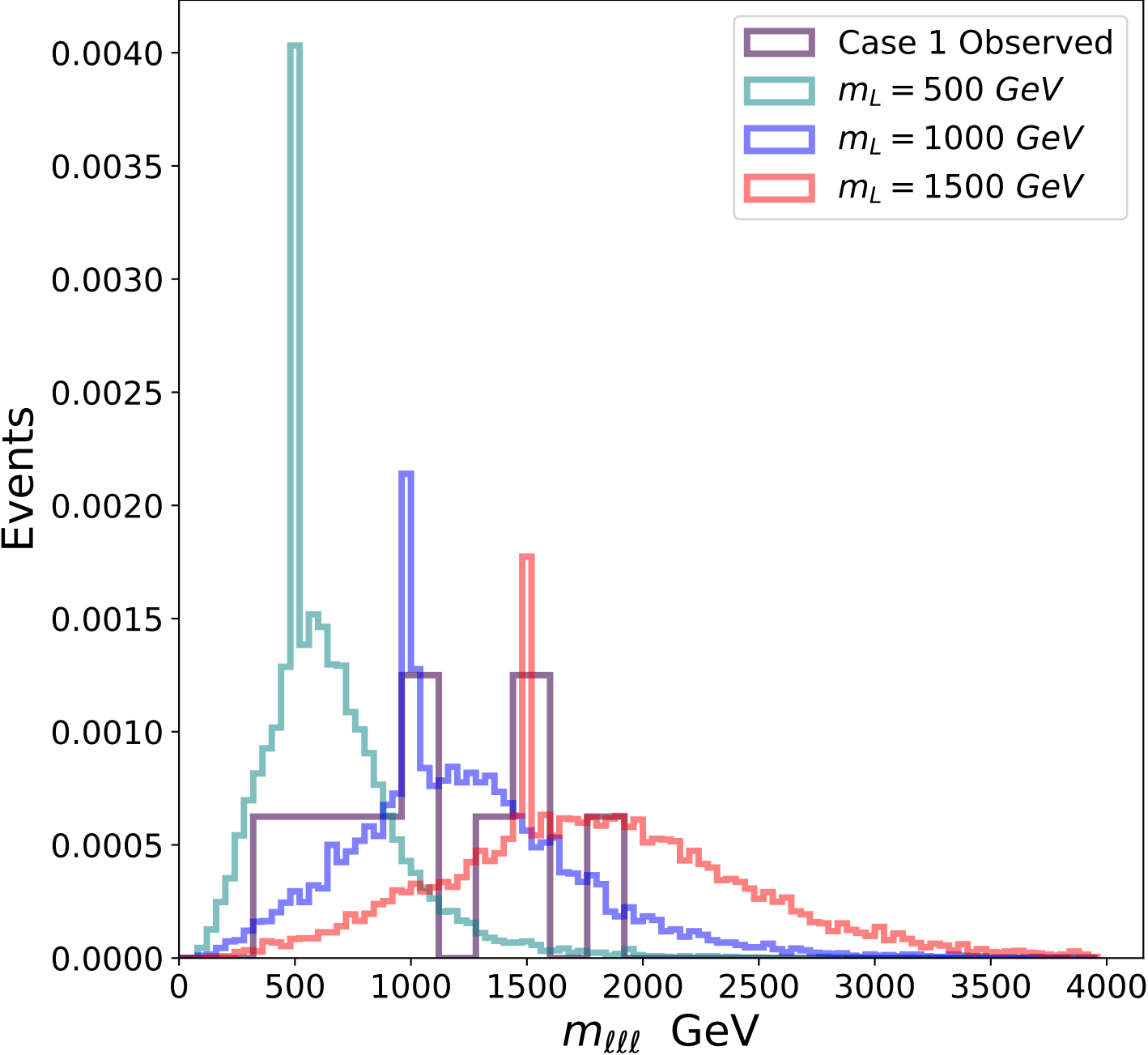
Label	SS500	SS1000	SS1500	SSSM
0	1.0	-1.000	1.000	-1.0	-1.0
1	1.0	-1.000	1.000	-1.0	-1.0
2	1.0	-1.000	1.000	-1.0	-1.0
3	1.0	-1.000	-1.000	-1.0	-1.0
4	1.0	-0.400	0.400	-1.0	-1.0
5	1.0	0.000	0.000	-1.0	-1.0
6	1.0	0.333	-0.333	-1.0	-1.0
7	1.0	-1.000	1.000	-1.0	-1.0
8	1.0	-1.000	0.000	0.0	-1.0
9	1.0	1.000	-1.000	-1.0	-1.0

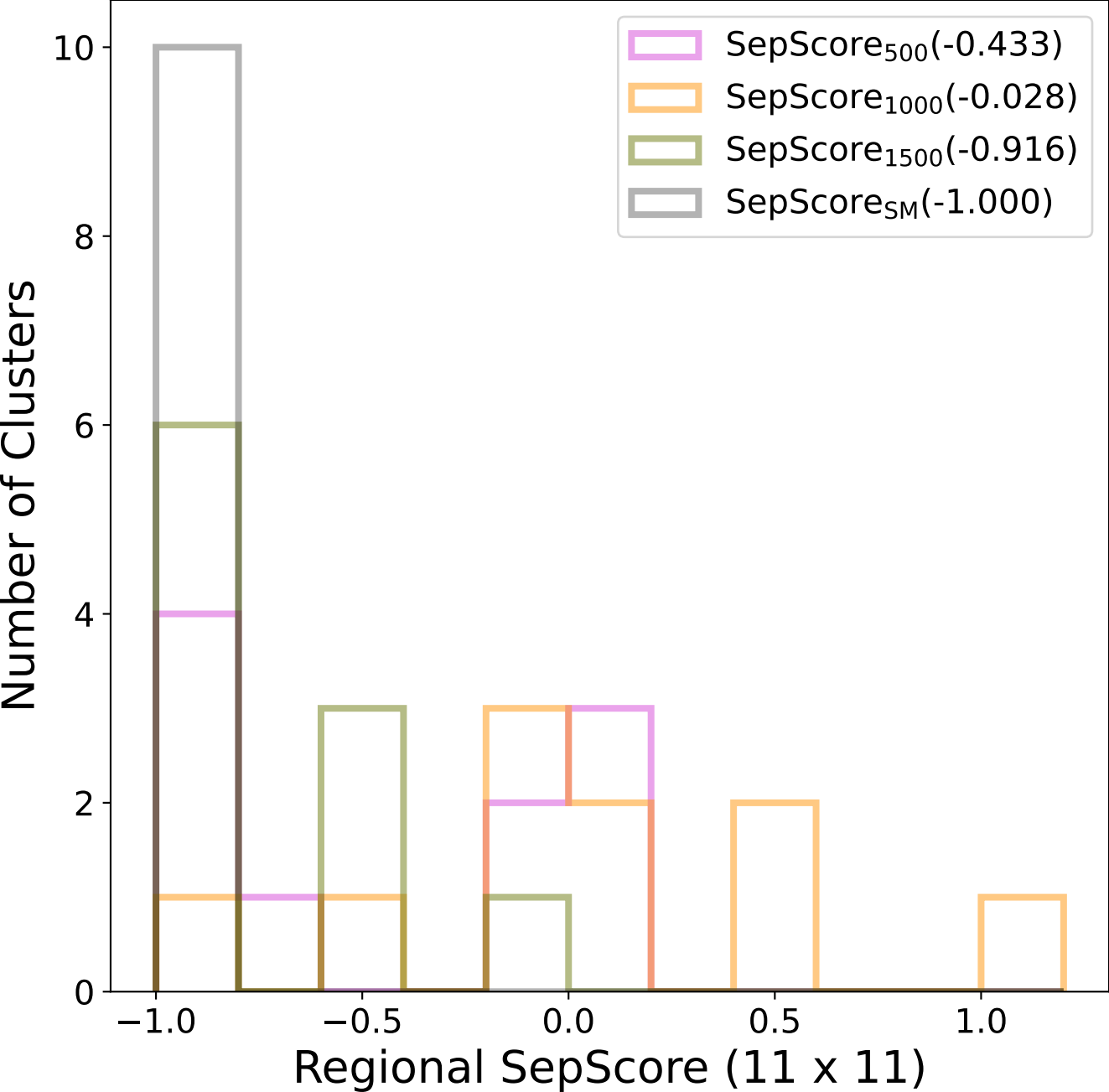
Array of label and scores AFTER ssSM < 0.6 cut: [label, ss500, ss1000, ss1500, ssSM]:

Label	SS500	SS1000	SS1500	SSSM
0	1.0	-1.000	1.000	-1.0	-1.0
1	1.0	-1.000	1.000	-1.0	-1.0
2	1.0	-1.000	1.000	-1.0	-1.0
3	1.0	-1.000	-1.000	-1.0	-1.0
4	1.0	-0.400	0.400	-1.0	-1.0
5	1.0	0.000	0.000	-1.0	-1.0
6	1.0	0.333	-0.333	-1.0	-1.0
7	1.0	-1.000	1.000	-1.0	-1.0
8	1.0	-1.000	0.000	0.0	-1.0
9	1.0	1.000	-1.000	-1.0	-1.0

Array of scores BEFORE ssSM < 0.6 cut: 10
Array of scores AFTER ssSM < 0.6 cut: 10


SepScore500(-1.000)
SepScore1000(0.200)
SepScore1500(-1.000)
SepScoreSM(-1.000)
Number of Clusters
0
2
4
6
8
10
−1.0
−0.5
0.0
0.5
1.0
Regional SepScore (5 x 5)


Case 1 Observed
$m_L = 500\ GeV$
$m_L = 1000\ GeV$
$m_L = 1500\ GeV$
Events
0.0040
0.0035
0.0030
0.0025
0.0020
0.0015
0.0010
0.0005
0.0000
0
500
1000
1500
2000
2500
3000
3500
4000
$m_{\ell\ell\ell}$ GeV


SepScore500(-0.433)
SepScore1000(-0.028)
SepScore1500(-0.916)
SepScoreSM(-1.000)
Number of Clusters
0
2
4
6
8
10
−1.0
−0.5
0.0
0.5
1.0
Regional SepScore (11 x 11)

Case 1: cluster of +- 5 around the point

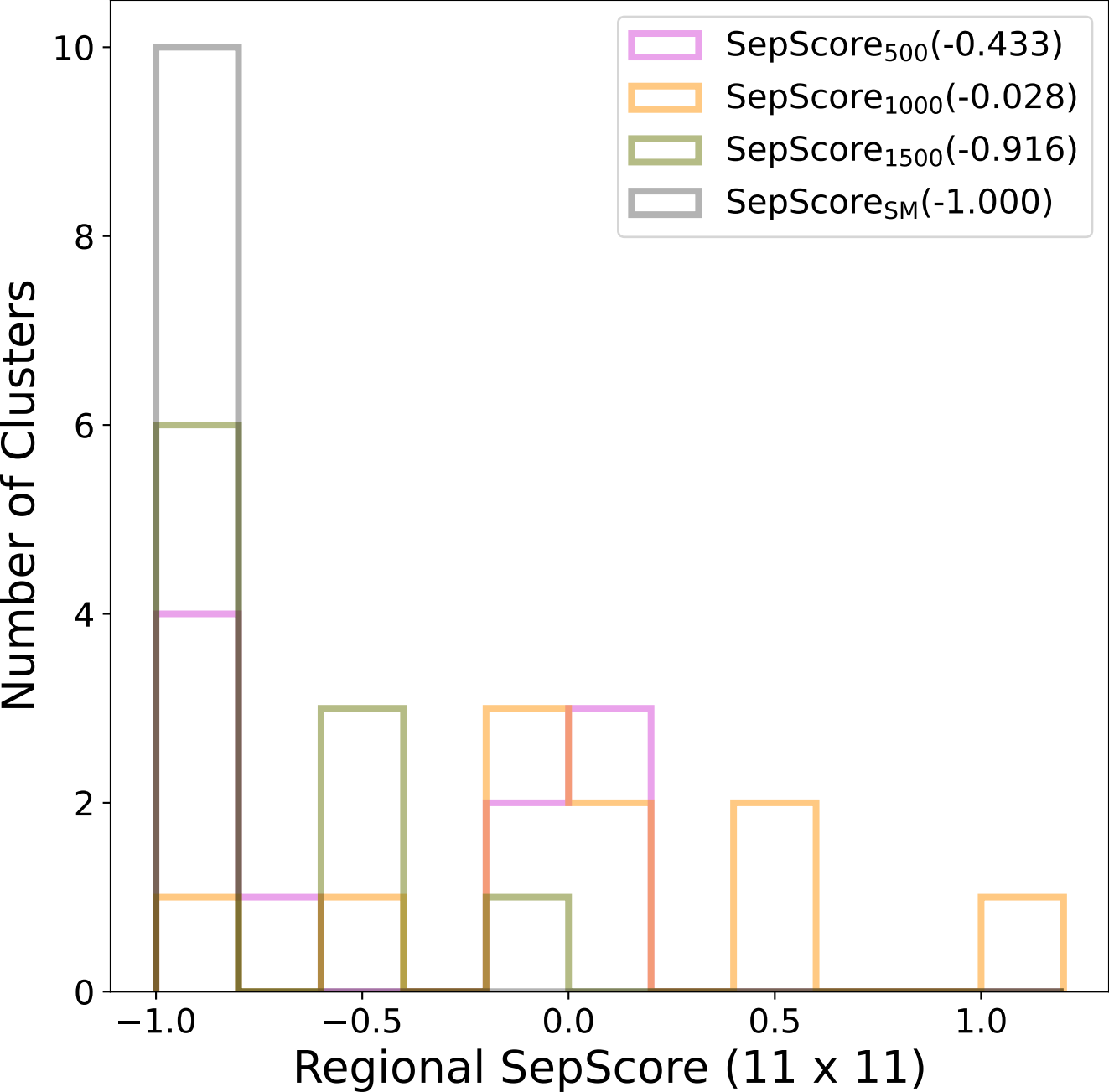
Array of label and scores BEFORE ssSM < 0.6 cut: [label, ss500, ss1000, ss1500, ssSM]:

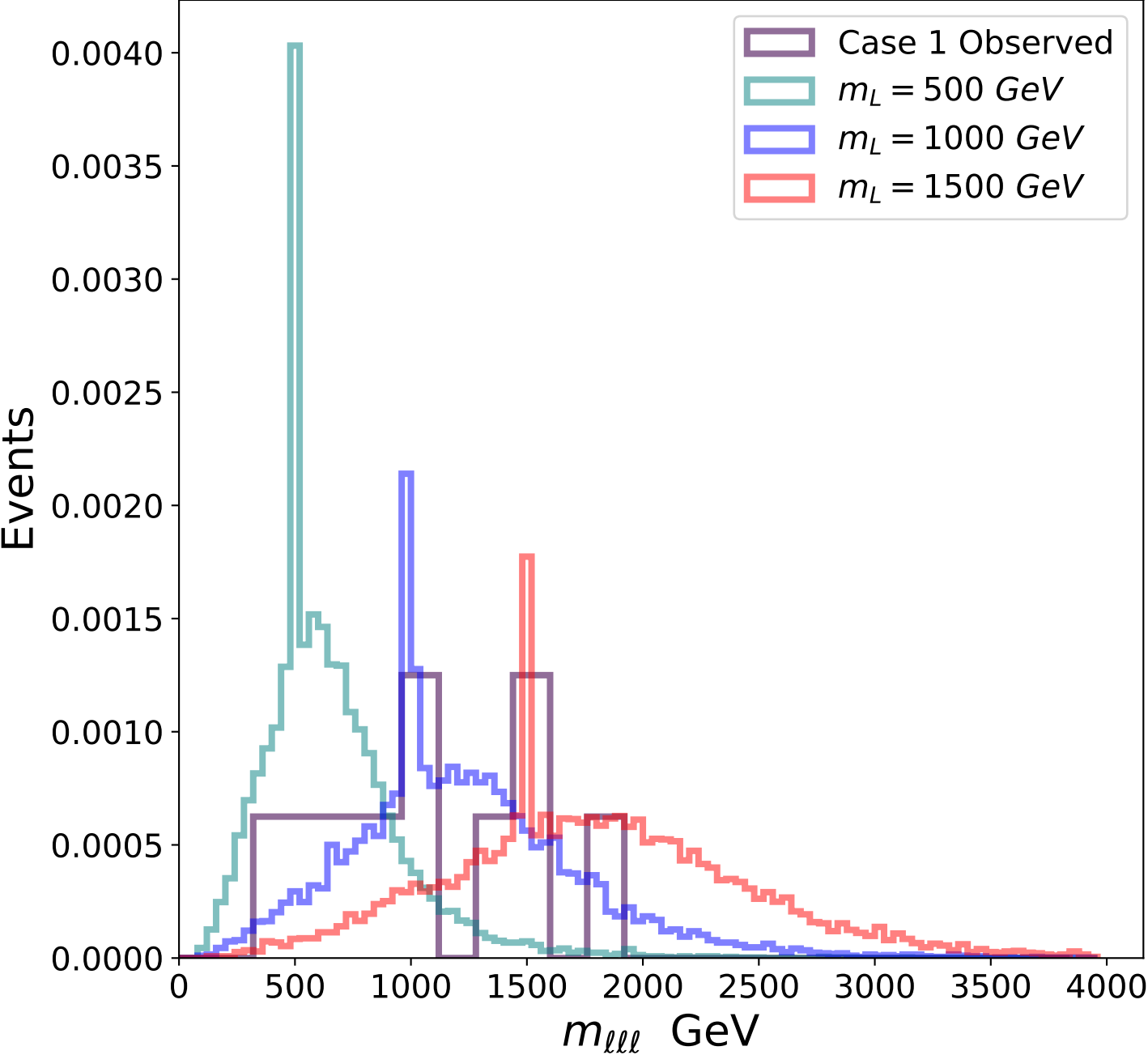
	Label	SS500	SS1000	SS1500	SSSM
0	1.0	-1.000	0.111	-0.111	-1.0
1	1.0	-1.000	0.500	-0.500	-1.0
2	1.0	-0.200	-0.200	-0.600	-1.0
3	1.0	0.000	-0.500	-0.500	-1.0
4	1.0	-0.140	0.140	-1.000	-1.0
5	1.0	-0.667	0.500	-0.833	-1.0
6	1.0	0.167	-0.167	-1.000	-1.0
7	1.0	-1.000	1.000	-1.000	-1.0
8	1.0	-1.000	-1.000	-1.000	-1.0
9	1.0	0.167	-0.167	-1.000	-1.0

Array of label and scores AFTER ssSM < 0.6 cut: [label, ss500, ss1000, ss1500, ssSM]:

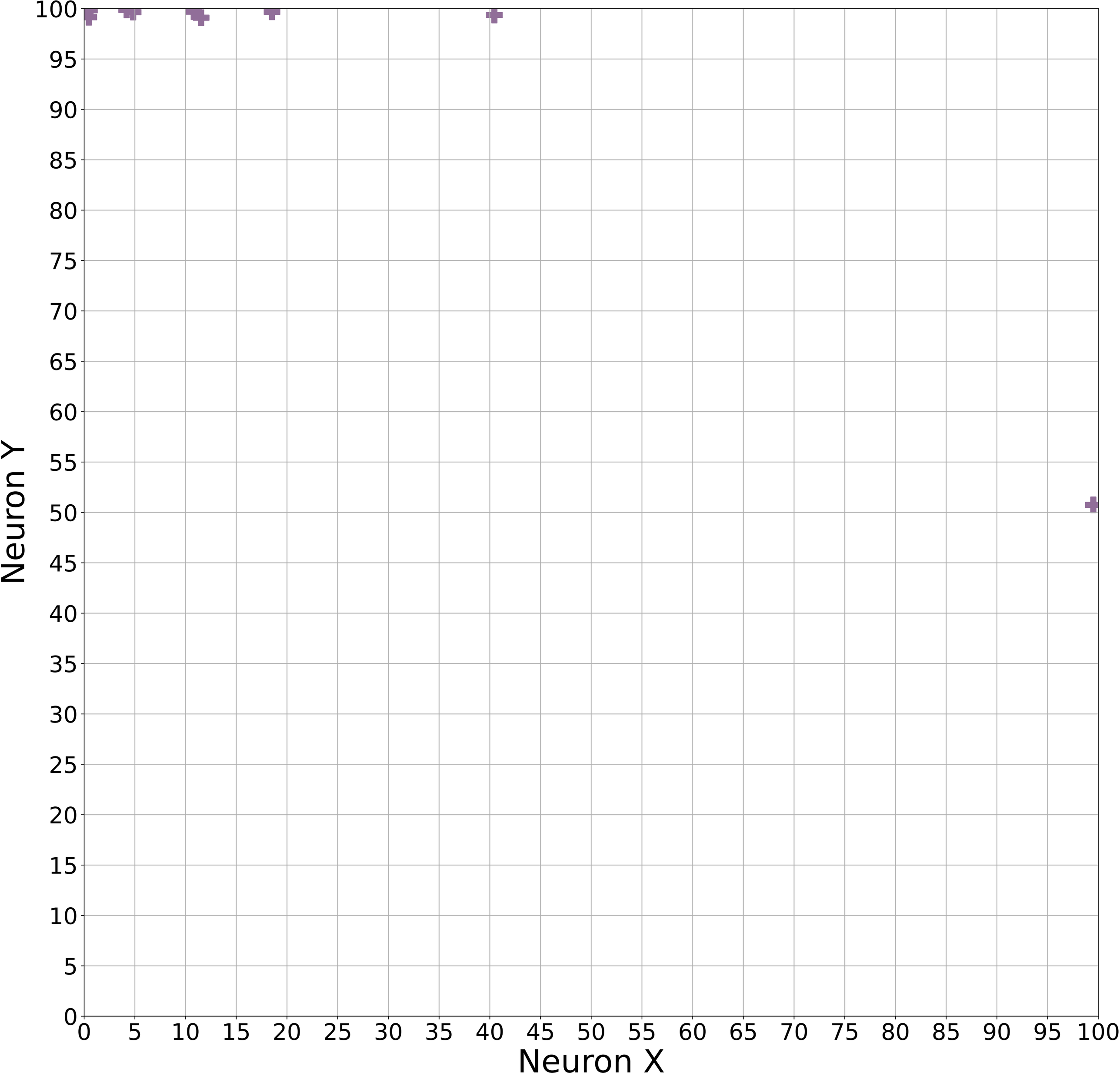
	Label	SS500	SS1000	SS1500	SSSM
0	1.0	-1.000	0.111	-0.111	-1.0
1	1.0	-1.000	0.500	-0.500	-1.0
2	1.0	-0.200	-0.200	-0.600	-1.0
3	1.0	0.000	-0.500	-0.500	-1.0
4	1.0	-0.140	0.140	-1.000	-1.0
5	1.0	-0.667	0.500	-0.833	-1.0
6	1.0	0.167	-0.167	-1.000	-1.0
7	1.0	-1.000	1.000	-1.000	-1.0
8	1.0	-1.000	-1.000	-1.000	-1.0
9	1.0	0.167	-0.167	-1.000	-1.0

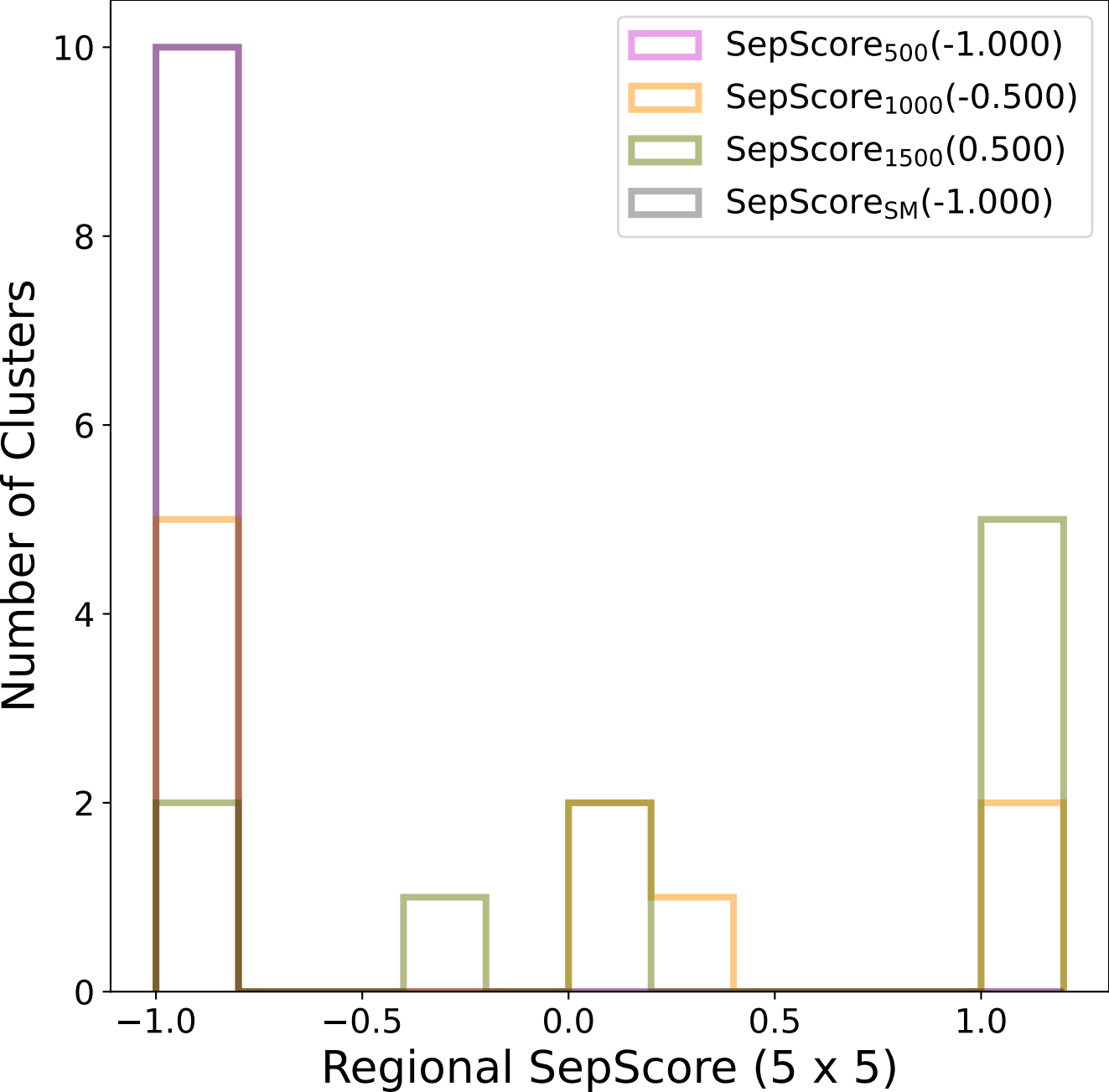
Array of scores BEFORE ssSM < 0.6 cut: 10
Array of scores AFTER ssSM < 0.6 cut: 10


SepScore500(-0.433)
SepScore1000(-0.028)
SepScore1500(-0.916)
SepScoreSM(-1.000)
Number of Clusters
0
2
4
6
8
10
−1.0
−0.5
0.0
0.5
1.0
Regional SepScore (11 x 11)


Case 1 Observed
$m_L = 500\ GeV$
$m_L = 1000\ GeV$
$m_L = 1500\ GeV$
Events
0.0040
0.0035
0.0030
0.0025
0.0020
0.0015
0.0010
0.0005
0.0000
0
500
1000
1500
2000
2500
3000
3500
4000
$m_{\ell\ell\ell}$ GeV

Case 2


Neuron Y
Neuron X
0
5
10
15
20
25
30
35
40
45
50
55
60
65
70
75
80
85
90
95
100


SepScore500(-1.000)
SepScore1000(-0.500)
SepScore1500(0.500)
SepScoreSM(-1.000)
Number of Clusters
0
2
4
6
8
10
−1.0
−0.5
0.0
0.5
1.0
Regional SepScore (5 x 5)

Case 2: cluster of +- 2 around the point

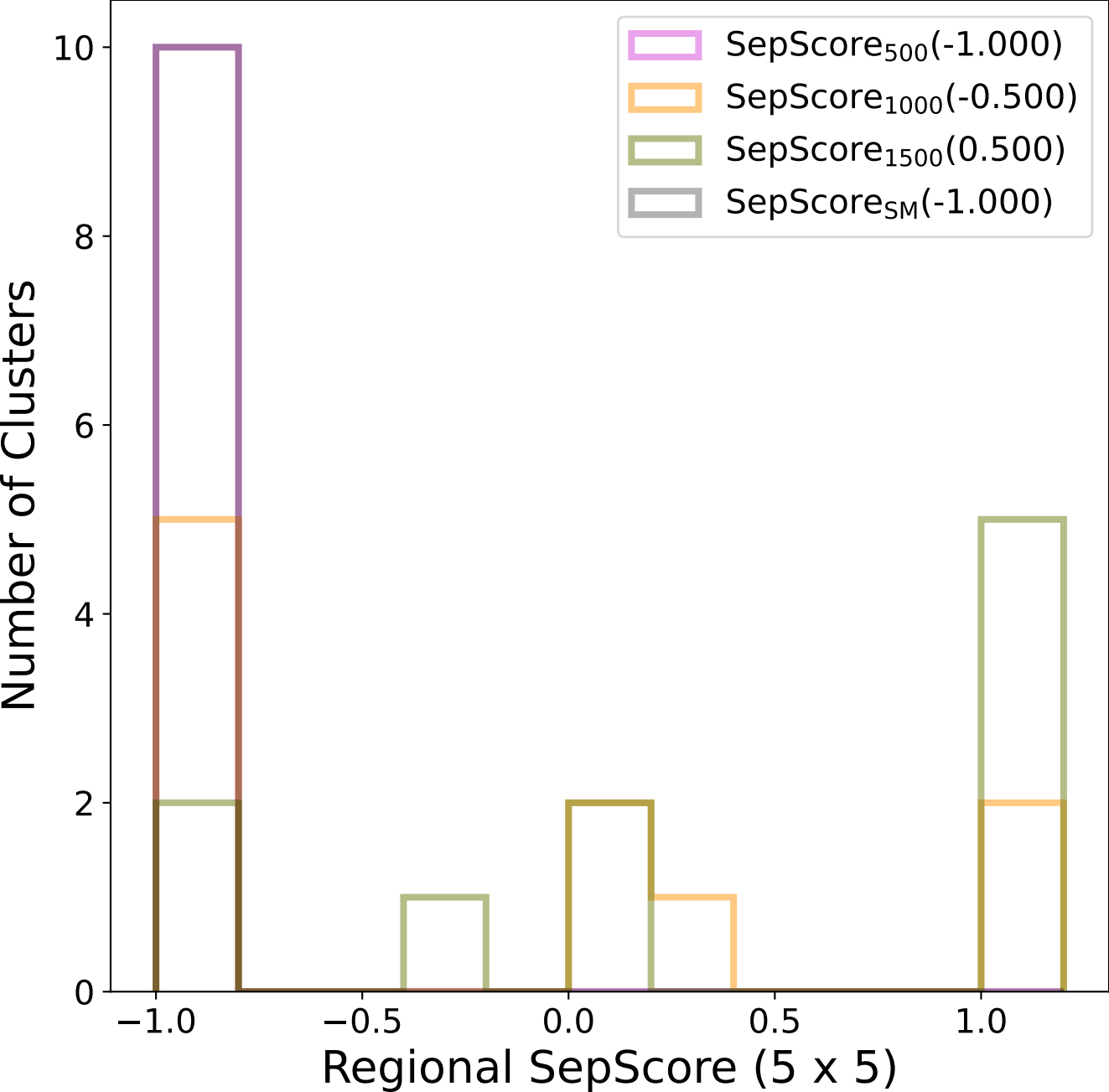
Array of label and scores BEFORE ssSM < 0.6 cut: [label, ss500, ss1000, ss1500, ssSM]:

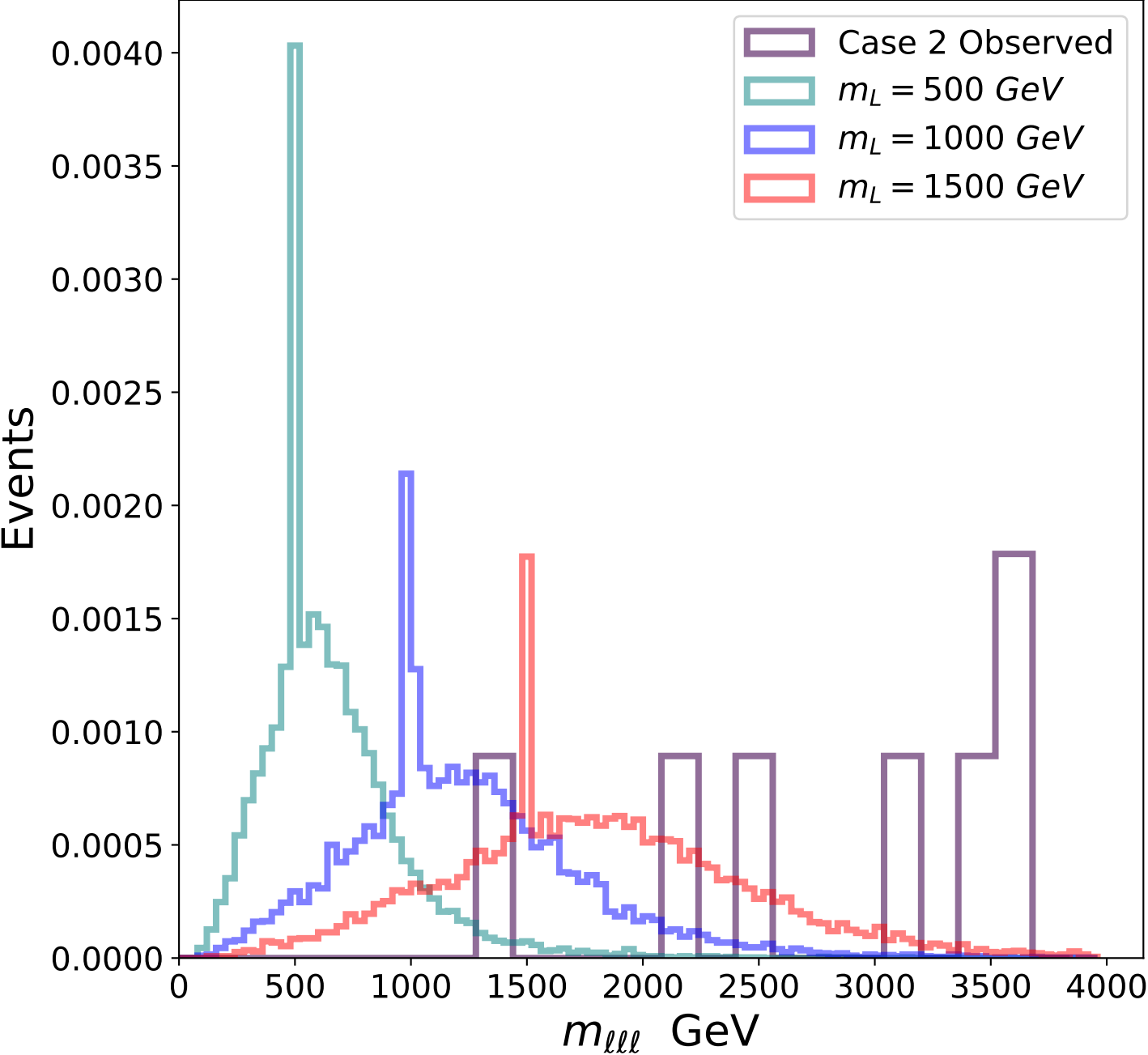
Label	SS500	SS1000	SS1500	SSSM
0	5.0	-1.0	0.000	0.000	-1.0
1	5.0	-1.0	0.000	0.000	-1.0
2	5.0	-1.0	1.000	-1.000	-1.0
3	5.0	-1.0	-1.000	1.000	-1.0
4	5.0	-1.0	0.333	-0.333	-1.0
5	5.0	-1.0	-1.000	1.000	-1.0
6	5.0	-1.0	-1.000	1.000	-1.0
7	5.0	-1.0	-1.000	1.000	-1.0
8	5.0	-1.0	-1.000	1.000	-1.0
9	5.0	-1.0	1.000	-1.000	-1.0

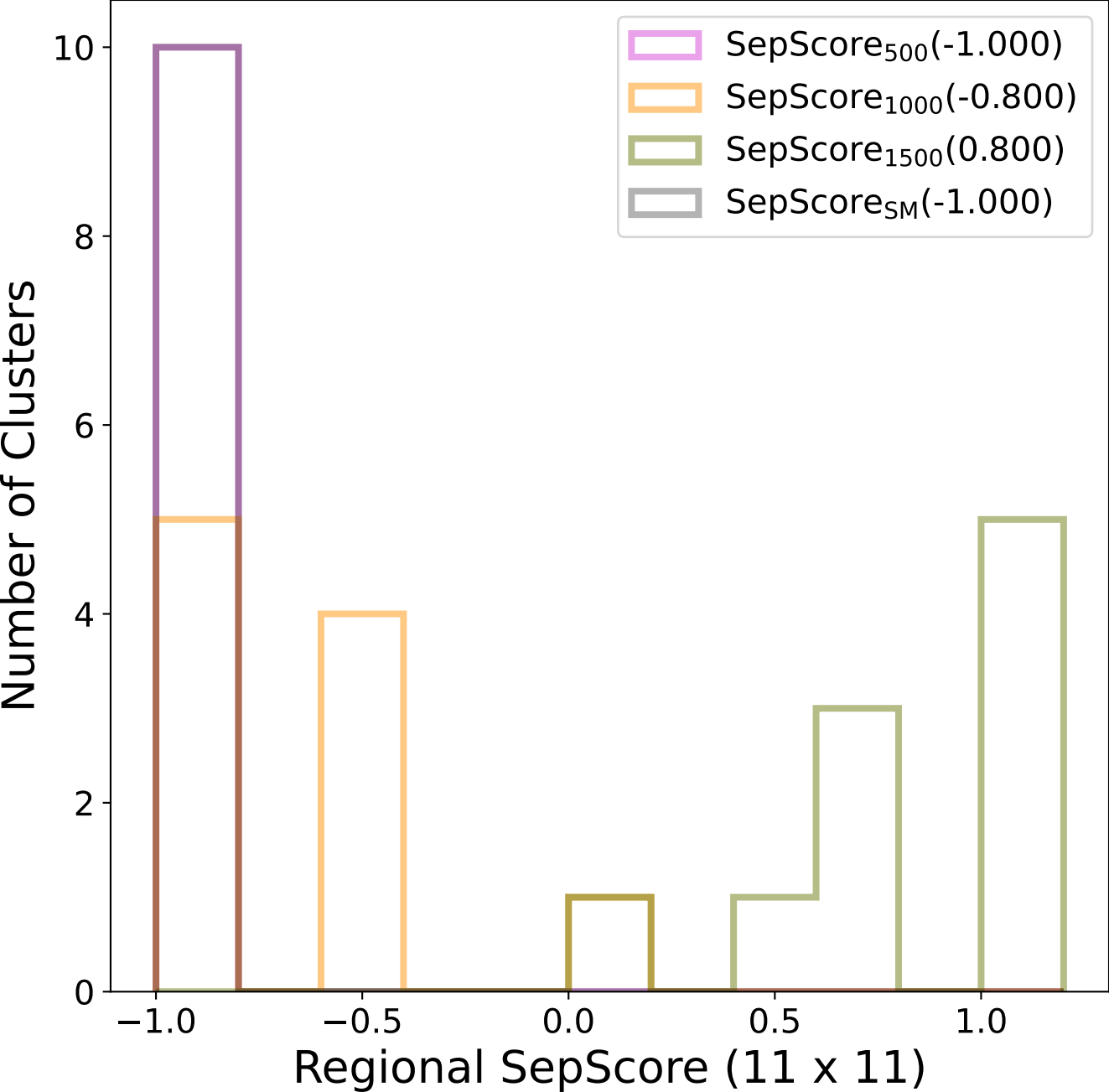
Array of label and scores AFTER ssSM < 0.6 cut: [label, ss500, ss1000, ss1500, ssSM]:

Label	SS500	SS1000	SS1500	SSSM
0	5.0	-1.0	0.000	0.000	-1.0
1	5.0	-1.0	0.000	0.000	-1.0
2	5.0	-1.0	1.000	-1.000	-1.0
3	5.0	-1.0	-1.000	1.000	-1.0
4	5.0	-1.0	0.333	-0.333	-1.0
5	5.0	-1.0	-1.000	1.000	-1.0
6	5.0	-1.0	-1.000	1.000	-1.0
7	5.0	-1.0	-1.000	1.000	-1.0
8	5.0	-1.0	-1.000	1.000	-1.0
9	5.0	-1.0	1.000	-1.000	-1.0

Array of scores BEFORE ssSM < 0.6 cut: 10
Array of scores AFTER ssSM < 0.6 cut: 10


SepScore500(-1.000)
SepScore1000(-0.500)
SepScore1500(0.500)
SepScoreSM(-1.000)
Number of Clusters
0
2
4
6
8
10
−1.0
−0.5
0.0
0.5
1.0
Regional SepScore (5 x 5)


Case 2 Observed
$m_L = 500\ GeV$
$m_L = 1000\ GeV$
$m_L = 1500\ GeV$
Events
0.0040
0.0035
0.0030
0.0025
0.0020
0.0015
0.0010
0.0005
0.0000
0
500
1000
1500
2000
2500
3000
3500
4000
$m_{\ell\ell\ell}$ GeV


SepScore500(-1.000)
SepScore1000(-0.800)
SepScore1500(0.800)
SepScoreSM(-1.000)
Number of Clusters
0
2
4
6
8
10
−1.0
−0.5
0.0
0.5
1.0
Regional SepScore (11 x 11)

Case 2: cluster of +- 5 around the point

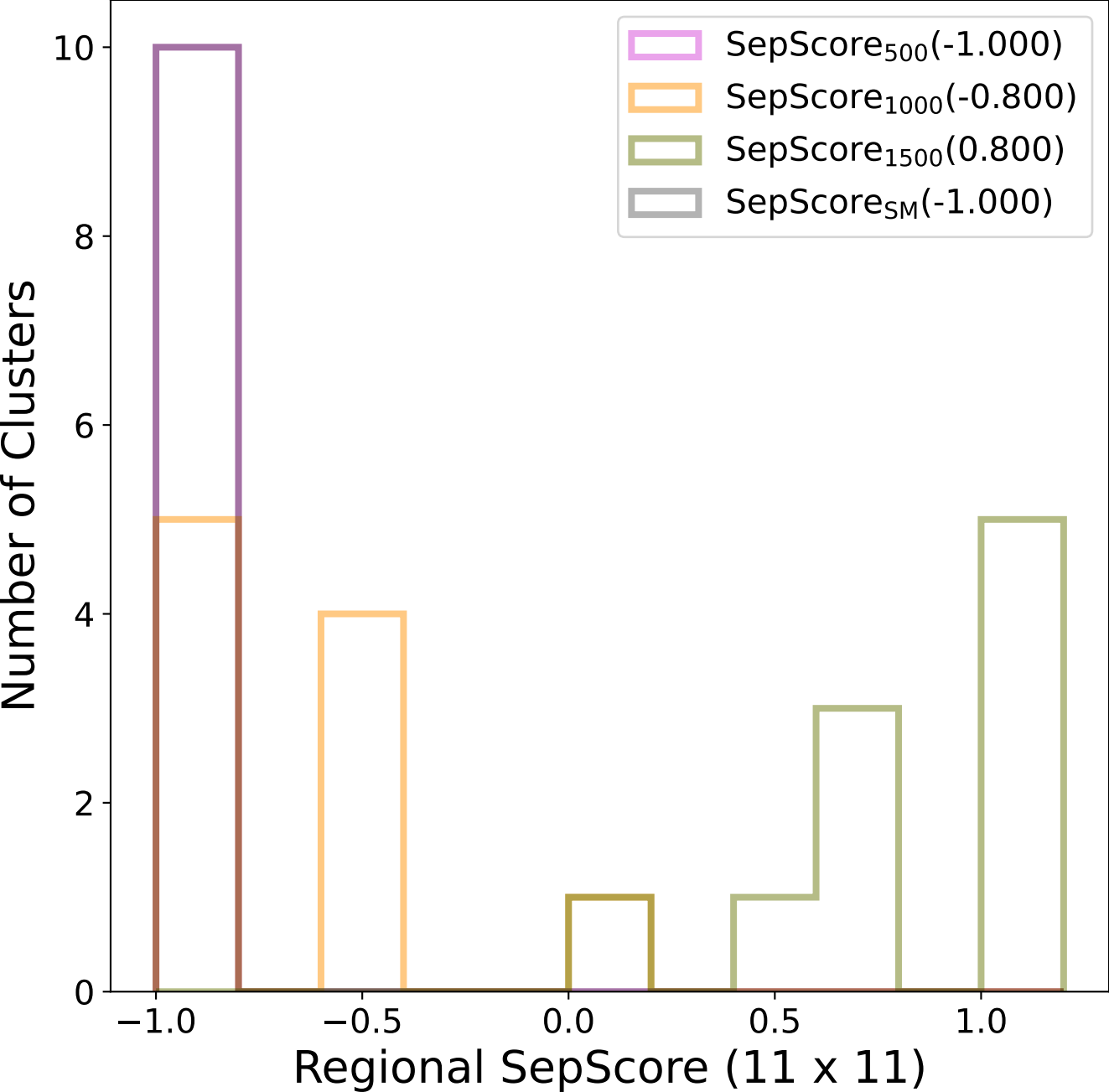
Array of label and scores BEFORE ssSM < 0.6 cut: [label, ss500, ss1000, ss1500, ssSM]:

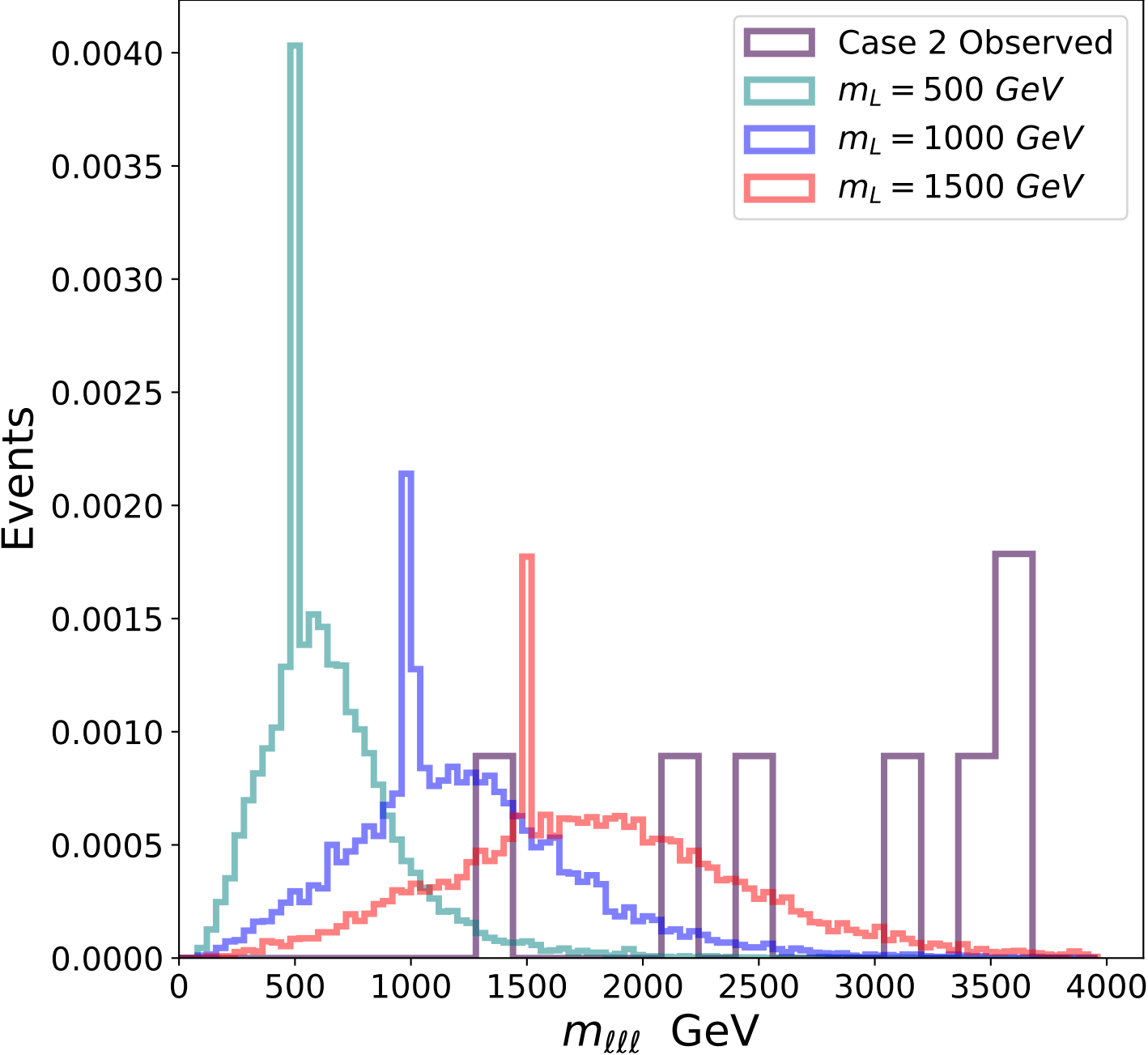
Label	SS500	SS1000	SS1500	SSSM
0	5.0	-1.0	-0.5	0.5	-1.0
1	5.0	-1.0	-0.6	0.6	-1.0
2	5.0	-1.0	-0.6	0.6	-1.0
3	5.0	-1.0	-1.0	1.0	-1.0
4	5.0	-1.0	0.0	0.0	-1.0
5	5.0	-1.0	-1.0	1.0	-1.0
6	5.0	-1.0	-1.0	1.0	-1.0
7	5.0	-1.0	-1.0	1.0	-1.0
8	5.0	-1.0	-1.0	1.0	-1.0
9	5.0	-1.0	-0.6	0.6	-1.0

Array of label and scores AFTER ssSM < 0.6 cut: [label, ss500, ss1000, ss1500, ssSM]:

Label	SS500	SS1000	SS1500	SSSM
0	5.0	-1.0	-0.5	0.5	-1.0
1	5.0	-1.0	-0.6	0.6	-1.0
2	5.0	-1.0	-0.6	0.6	-1.0
3	5.0	-1.0	-1.0	1.0	-1.0
4	5.0	-1.0	0.0	0.0	-1.0
5	5.0	-1.0	-1.0	1.0	-1.0
6	5.0	-1.0	-1.0	1.0	-1.0
7	5.0	-1.0	-1.0	1.0	-1.0
8	5.0	-1.0	-1.0	1.0	-1.0
9	5.0	-1.0	-0.6	0.6	-1.0

Array of scores BEFORE ssSM < 0.6 cut: 10
Array of scores AFTER ssSM < 0.6 cut: 10


SepScore500(-1.000)
SepScore1000(-0.800)
SepScore1500(0.800)
SepScoreSM(-1.000)
Number of Clusters
0
2
4
6
8
10
−1.0
−0.5
0.0
0.5
1.0
Regional SepScore (11 x 11)


Case 2 Observed
$m_L = 500\ GeV$
$m_L = 1000\ GeV$
$m_L = 1500\ GeV$
Events
0.0040
0.0035
0.0030
0.0025
0.0020
0.0015
0.0010
0.0005
0.0000
0
500
1000
1500
2000
2500
3000
3500
4000
$m_{\ell\ell\ell}$ GeV

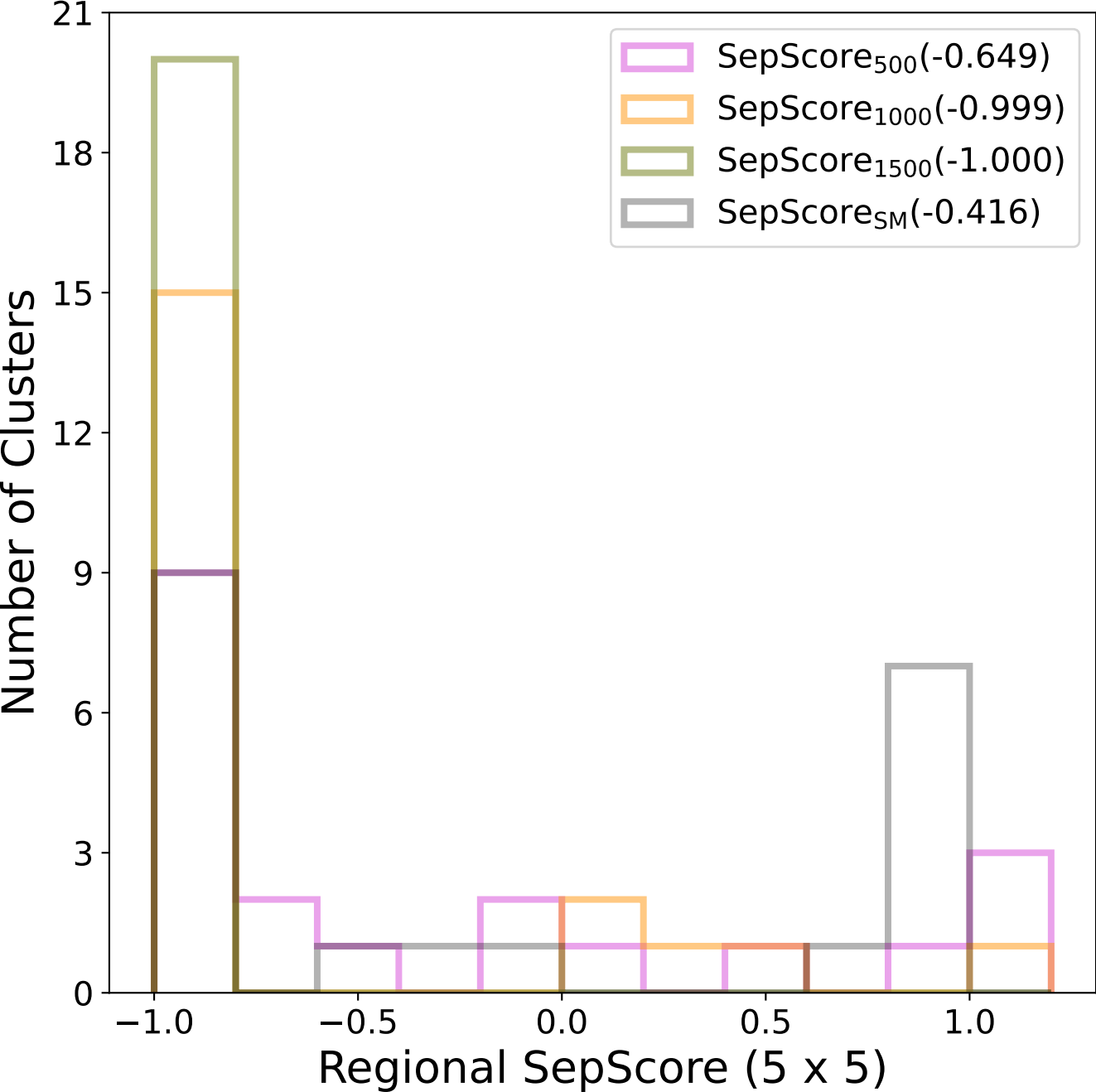
Case 3

Neuron Y

100
95
90
85
80
75
70
65
60
55
50
45
40
35
30
25
20
15
10
5
0

0 5 10 15 20 25 30 35 40 45 50 55 60 65 70 75 80 85 90 95 100

Neuron X


SepScore500(-0.649)
SepScore1000(-0.999)
SepScore1500(-1.000)
SepScoreSM(-0.416)
Number of Clusters
0
3
6
9
12
15
18
21
−1.0
−0.5
0.0
0.5
1.0
Regional SepScore (5 x 5)

Case 3: cluster of +- 2 around the point

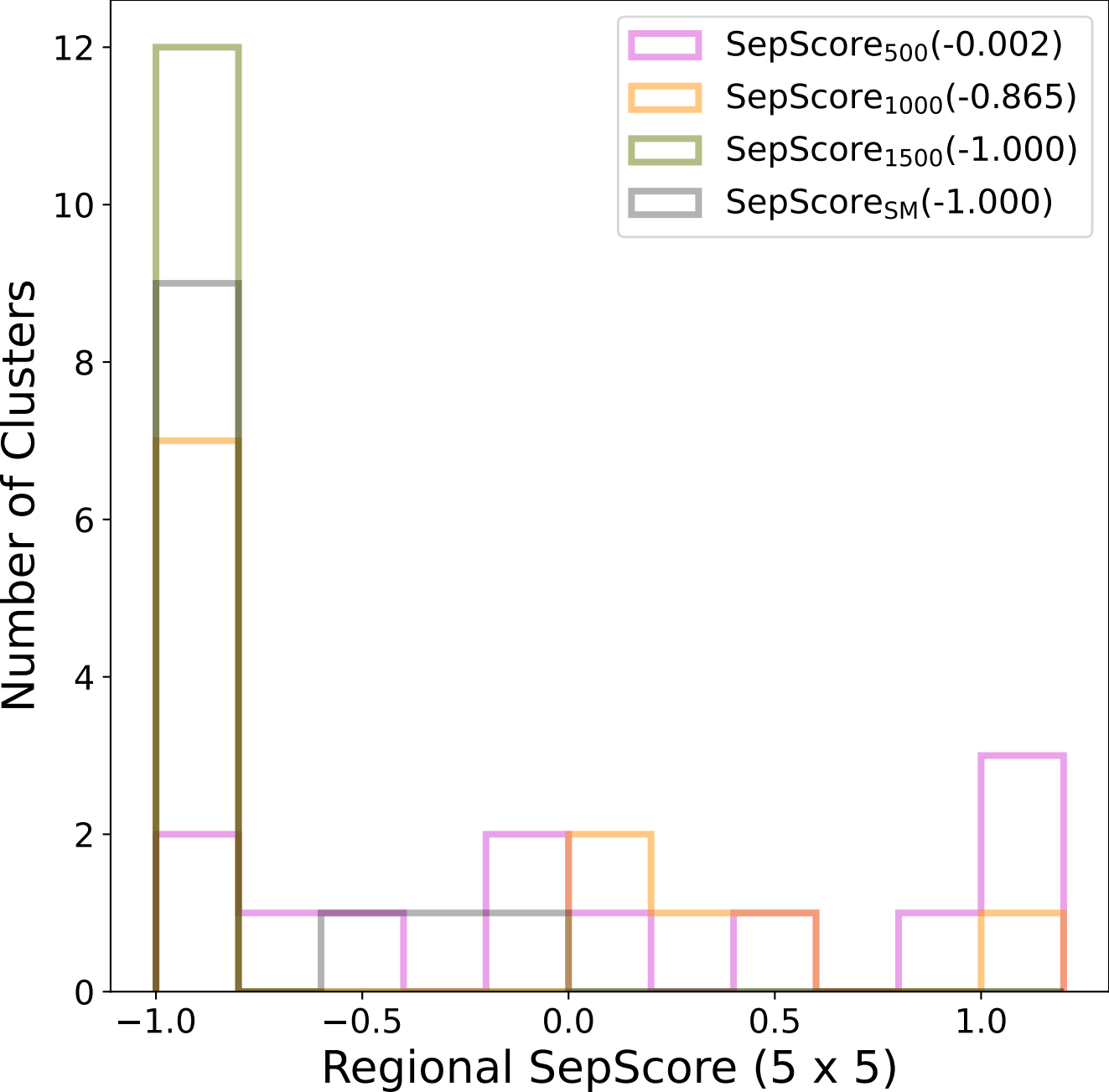
Array of label and scores BEFORE ssSM < 0.6 cut: [label, ss500, ss1000, ss1500, ssSM]:

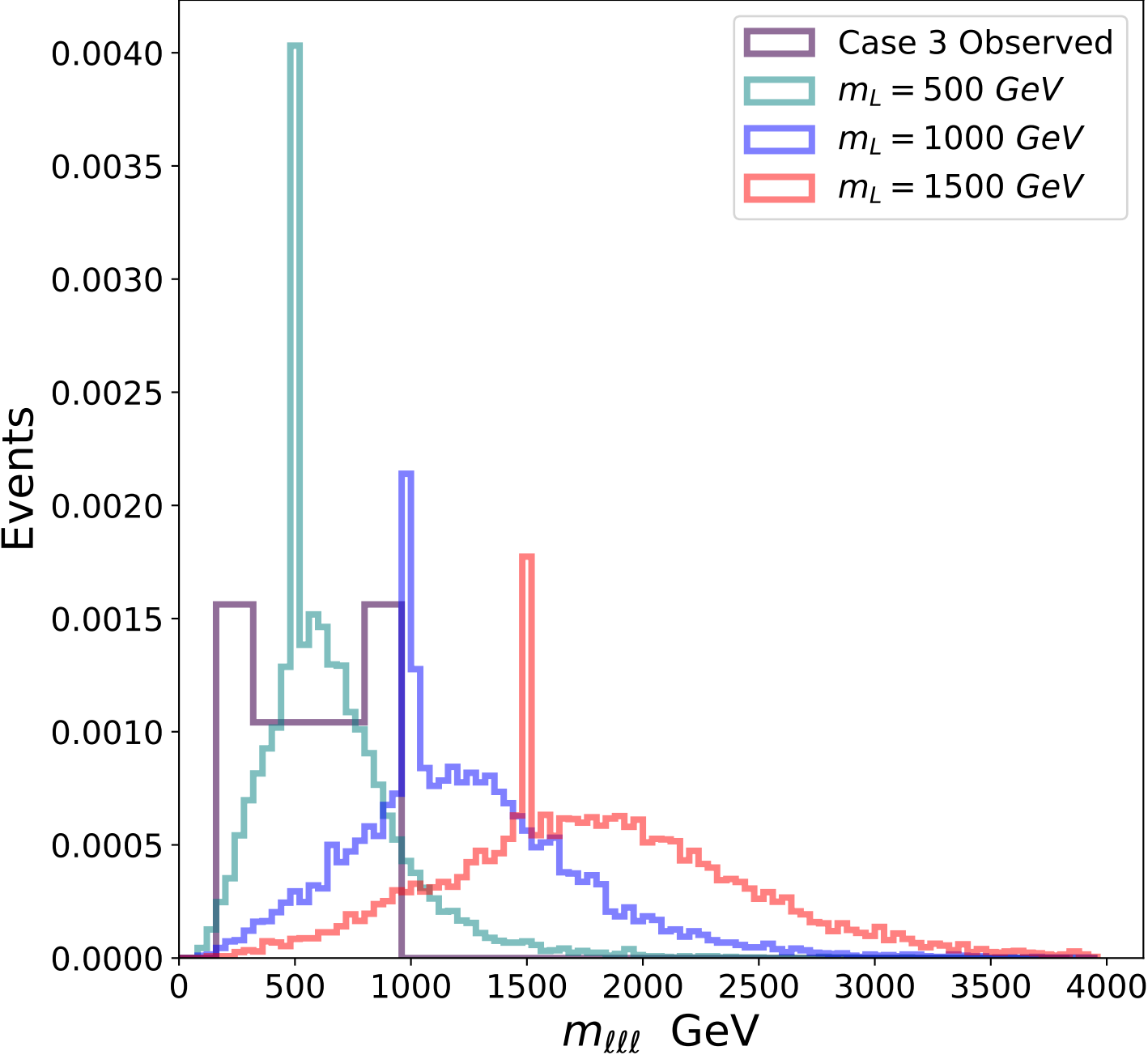
	Label	SS500	SS1000	SS1500	SSSM
0	4.0	-0.998	-1.000	-1.000	0.998
1	0.0	-0.200	0.200	-1.000	-1.000
2	0.0	-0.500	0.500	-1.000	-1.000
3	4.0	-0.881	-0.999	-1.000	0.880
4	4.0	-0.881	-0.999	-1.000	0.880
5	4.0	-0.881	-0.999	-1.000	0.880
6	0.0	0.840	-0.840	-1.000	-1.000
7	0.0	0.500	-1.000	-1.000	-0.500
8	4.0	-0.837	-0.999	-1.000	0.837
9	0.0	0.000	0.000	-1.000	-1.000
10	4.0	-0.003	-0.890	-0.973	-0.133
11	4.0	-0.881	-0.999	-1.000	0.880
12	0.0	1.000	-1.000	-1.000	-1.000
13	0.0	-1.000	1.000	-1.000	-1.000
14	0.0	1.000	-1.000	-1.000	-1.000
15	4.0	-0.881	-0.999	-1.000	0.880
16	4.0	-0.631	-1.000	-1.000	0.631
17	4.0	-1.000	-1.000	-1.000	-1.000
18	0.0	-0.667	0.000	-1.000	-0.333
19	0.0	1.000	-1.000	-1.000	-1.000

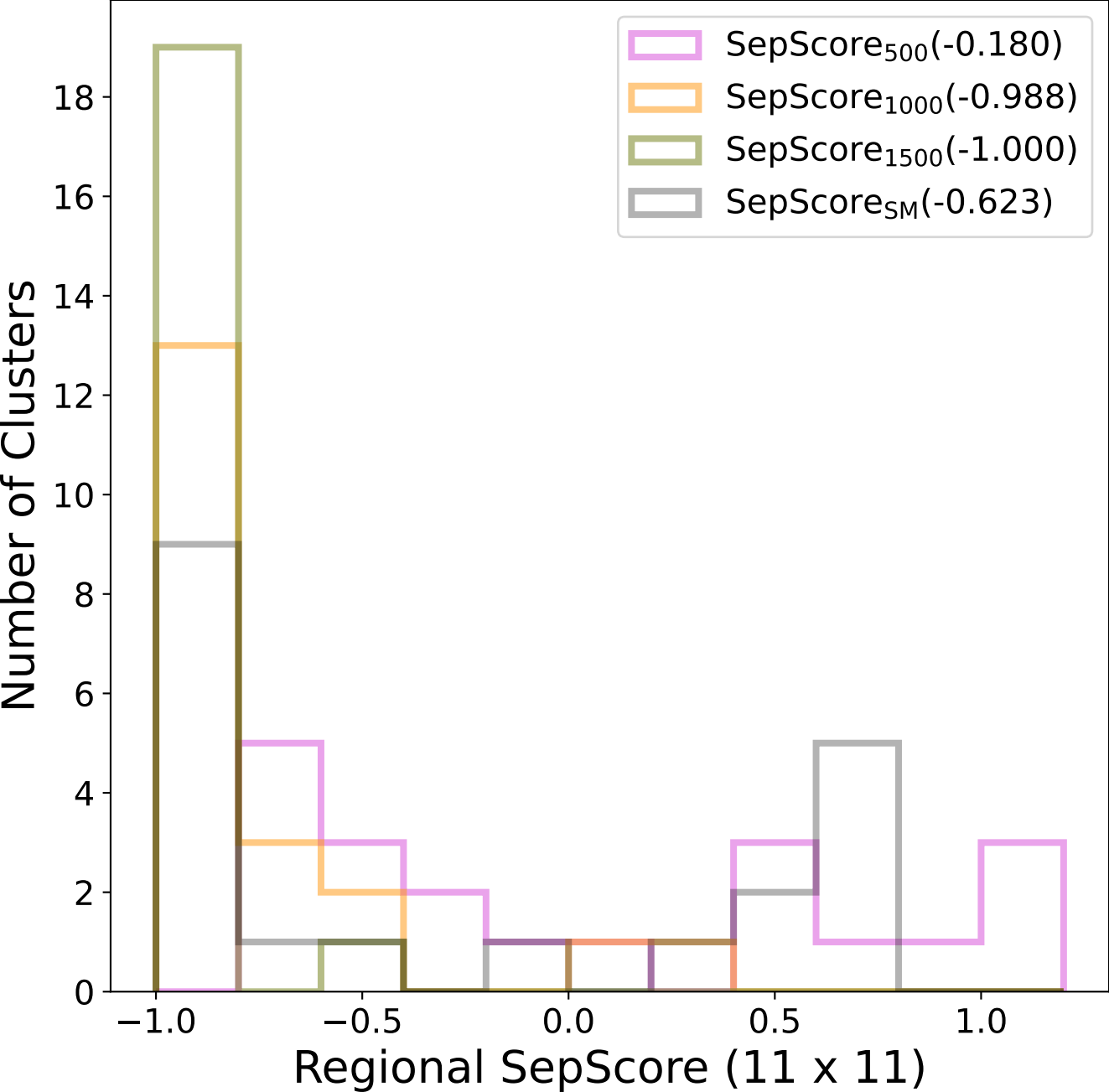
Array of label and scores AFTER ssSM < 0.6 cut: [label, ss500, ss1000, ss1500, ssSM]:

	Label	SS500	SS1000	SS1500	SSSM
1	0.0	-0.200	0.20	-1.000	-1.000
2	0.0	-0.500	0.50	-1.000	-1.000
6	0.0	0.840	-0.84	-1.000	-1.000
7	0.0	0.500	-1.00	-1.000	-0.500
9	0.0	0.000	0.00	-1.000	-1.000
10	4.0	-0.003	-0.89	-0.973	-0.133
12	0.0	1.000	-1.00	-1.000	-1.000
13	0.0	-1.000	1.00	-1.000	-1.000
14	0.0	1.000	-1.00	-1.000	-1.000
17	4.0	-1.000	-1.00	-1.000	-1.000
18	0.0	-0.667	0.00	-1.000	-0.333
19	0.0	1.000	-1.00	-1.000	-1.000

Array of scores BEFORE ssSM < 0.6 cut: 20
Array of scores AFTER ssSM < 0.6 cut: 12


SepScore500(-0.002)
SepScore1000(-0.865)
SepScore1500(-1.000)
SepScoreSM(-1.000)
Number of Clusters
0
2
4
6
8
10
12
−1.0
−0.5
0.0
0.5
1.0
Regional SepScore (5 x 5)


Case 3 Observed
$m_L = 500\ GeV$
$m_L = 1000\ GeV$
$m_L = 1500\ GeV$
Events
0.0040
0.0035
0.0030
0.0025
0.0020
0.0015
0.0010
0.0005
0.0000
0
500
1000
1500
2000
2500
3000
3500
4000
$m_{\ell\ell\ell}$ GeV


SepScore500(-0.180)
SepScore1000(-0.988)
SepScore1500(-1.000)
SepScoreSM(-0.623)
Number of Clusters
0
2
4
6
8
10
12
14
16
18
−1.0
−0.5
0.0
0.5
1.0
Regional SepScore (11 x 11)

Case 3: cluster of +- 5 around the point

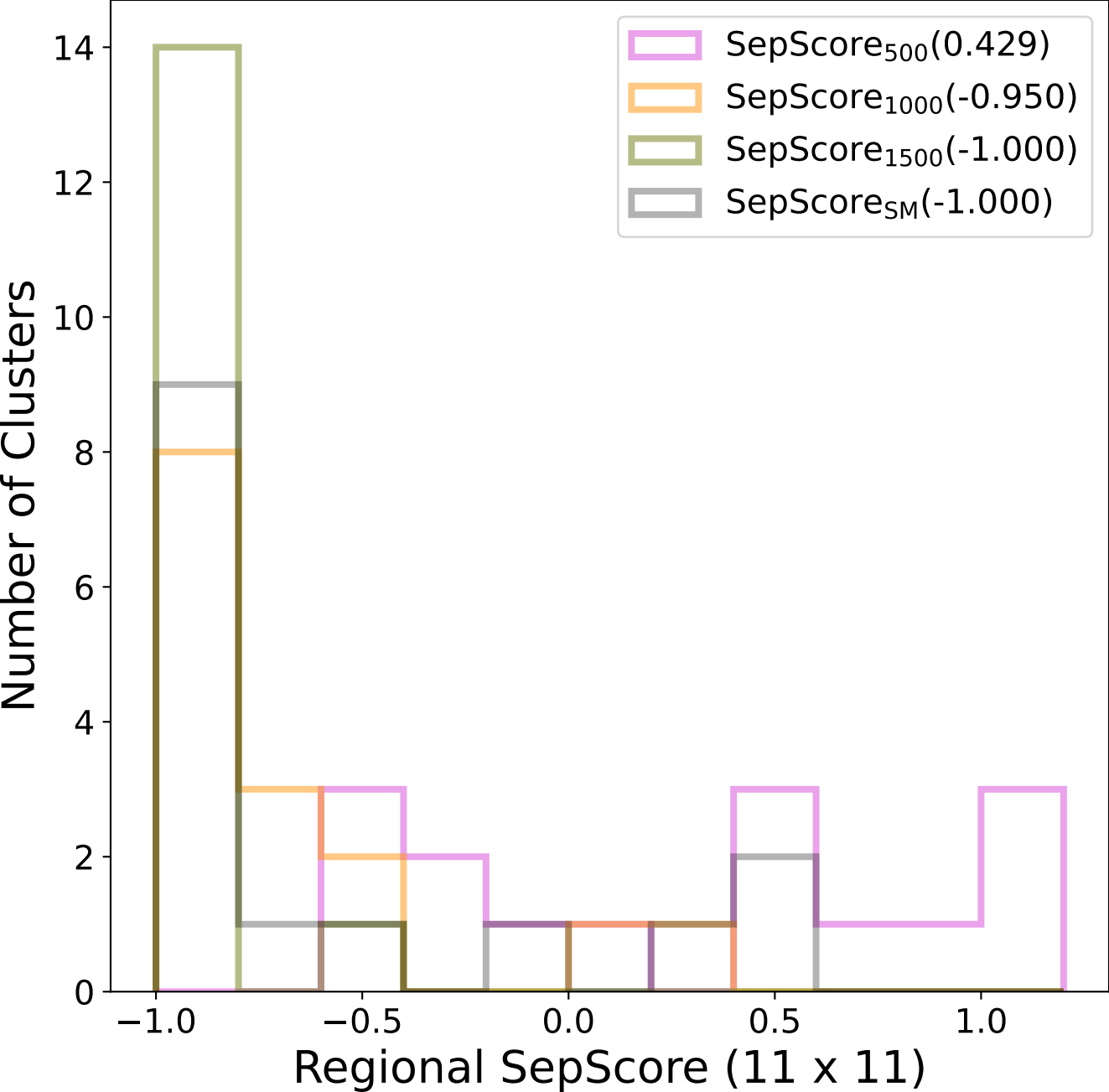
Array of label and scores BEFORE ssSM < 0.6 cut: [label, ss500, ss1000, ss1500, ssSM]:

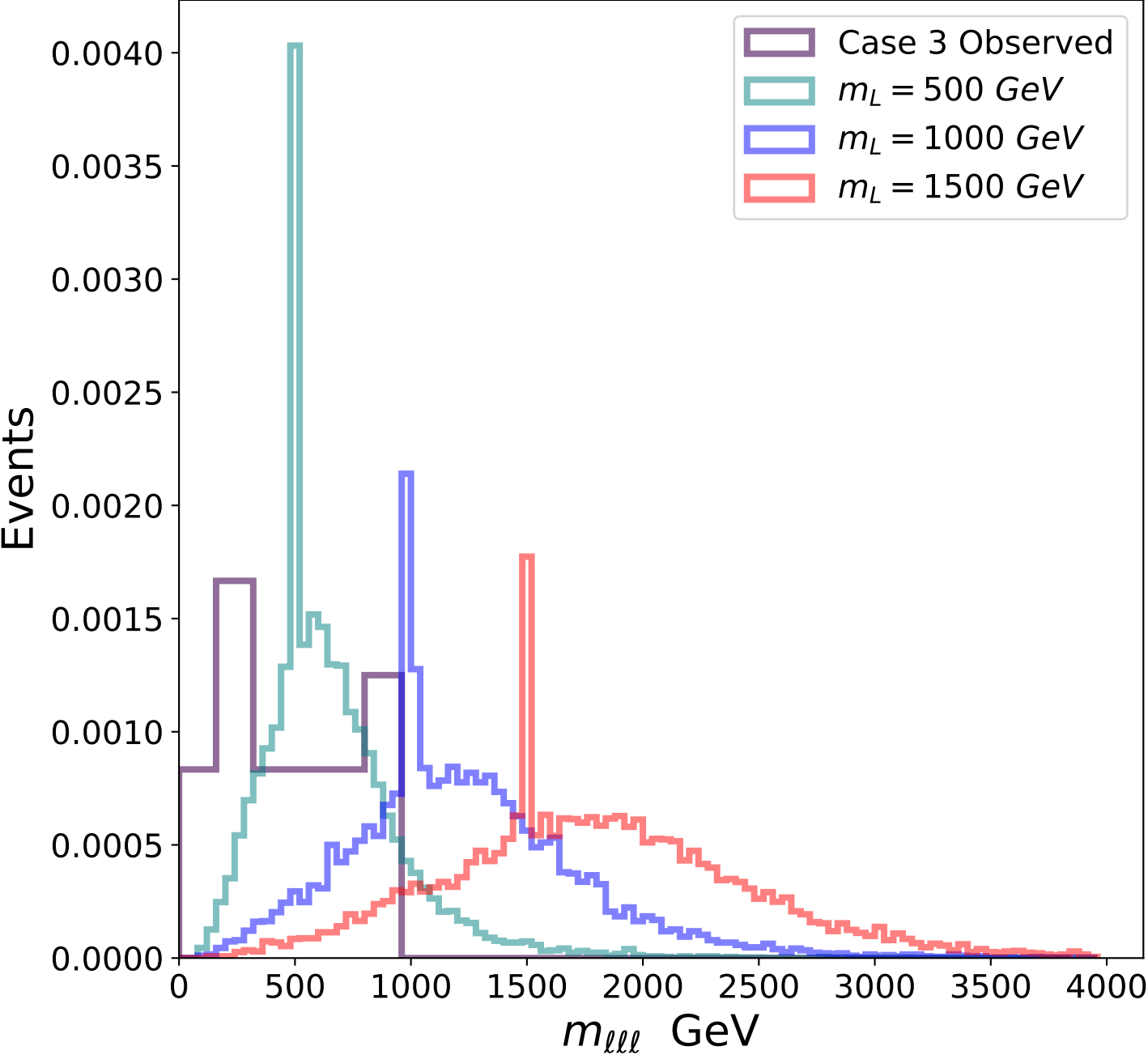
	Label	SS500	SS1000	SS1500	SSSM
0	4.0	-0.306	-1.000	-1.000	0.306
1	0.0	-0.350	0.300	-0.950	-1.000
2	0.0	-0.500	0.000	-0.500	-1.000
3	4.0	-0.616	-1.000	-1.000	0.616
4	4.0	-0.616	-1.000	-1.000	0.616
5	4.0	-0.616	-1.000	-1.000	0.616
6	0.0	0.553	-0.620	-0.933	-1.000
7	0.0	0.497	-1.000	-1.000	-0.497
8	4.0	-0.571	-0.977	-1.000	0.548
9	0.0	0.125	-0.438	-0.938	-0.750
10	4.0	-0.055	-0.950	-0.988	-0.008
11	4.0	-0.616	-1.000	-1.000	0.616
12	0.0	0.800	-0.800	-1.000	-1.000
13	0.0	0.429	-0.429	-1.000	-1.000
14	0.0	1.000	-1.000	-1.000	-1.000
15	4.0	-0.616	-1.000	-1.000	0.616
16	4.0	-0.524	-0.962	-1.000	0.485
17	4.0	1.000	-1.000	-1.000	-1.000
18	0.0	0.625	-0.750	-1.000	-0.875
19	0.0	1.000	-1.000	-1.000	-1.000

Array of label and scores AFTER ssSM < 0.6 cut: [label, ss500, ss1000, ss1500, ssSM]:

	Label	SS500	SS1000	SS1500	SSSM
0	4.0	-0.306	-1.000	-1.000	0.306
1	0.0	-0.350	0.300	-0.950	-1.000
2	0.0	-0.500	0.000	-0.500	-1.000
6	0.0	0.553	-0.620	-0.933	-1.000
7	0.0	0.497	-1.000	-1.000	-0.497
8	4.0	-0.571	-0.977	-1.000	0.548
9	0.0	0.125	-0.438	-0.938	-0.750
10	4.0	-0.055	-0.950	-0.988	-0.008
12	0.0	0.800	-0.800	-1.000	-1.000
13	0.0	0.429	-0.429	-1.000	-1.000
14	0.0	1.000	-1.000	-1.000	-1.000
16	4.0	-0.524	-0.962	-1.000	0.485
17	4.0	1.000	-1.000	-1.000	-1.000
18	0.0	0.625	-0.750	-1.000	-0.875
19	0.0	1.000	-1.000	-1.000	-1.000

Array of scores BEFORE ssSM < 0.6 cut: 20
Array of scores AFTER ssSM < 0.6 cut: 15


SepScore500(0.429)
SepScore1000(-0.950)
SepScore1500(-1.000)
SepScoreSM(-1.000)
Number of Clusters
0
2
4
6
8
10
12
14
−1.0
−0.5
0.0
0.5
1.0
Regional SepScore (11 x 11)


Case 3 Observed
$m_L = 500\ GeV$
$m_L = 1000\ GeV$
$m_L = 1500\ GeV$
Events
0.0040
0.0035
0.0030
0.0025
0.0020
0.0015
0.0010
0.0005
0.0000
0
500
1000
1500
2000
2500
3000
3500
4000
$m_{\ell\ell\ell}$ GeV

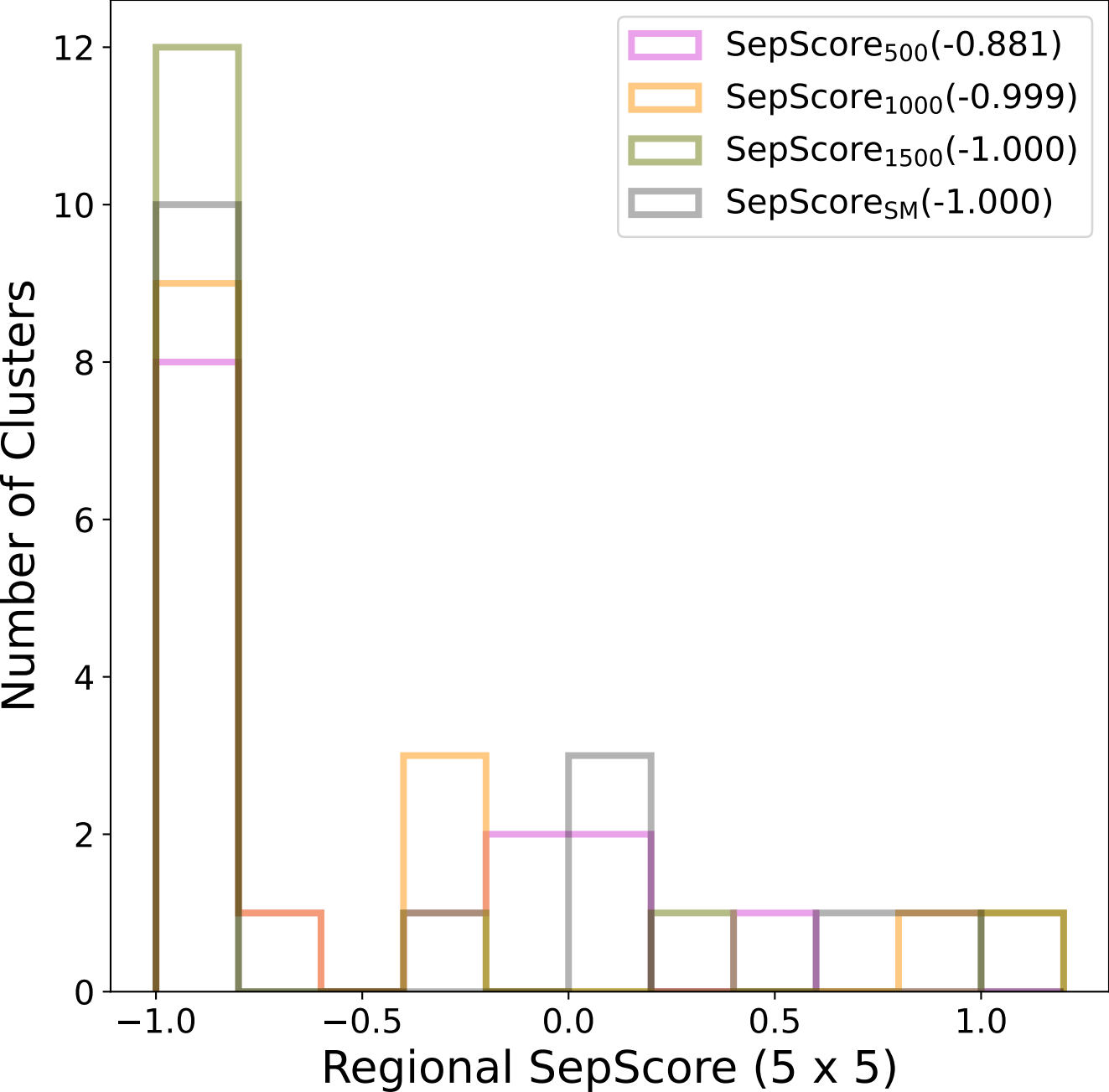
Case 4

Neuron Y

100
95
90
85
80
75
70
65
60
55
50
45
40
35
30
25
20
15
10
5
0

0 5 10 15 20 25 30 35 40 45 50 55 60 65 70 75 80 85 90 95 100

Neuron X


SepScore500(-0.881)
SepScore1000(-0.999)
SepScore1500(-1.000)
SepScoreSM(-1.000)
Number of Clusters
0
2
4
6
8
10
12
−1.0
−0.5
0.0
0.5
1.0
Regional SepScore (5 x 5)

Case 4: cluster of +- 2 around the point

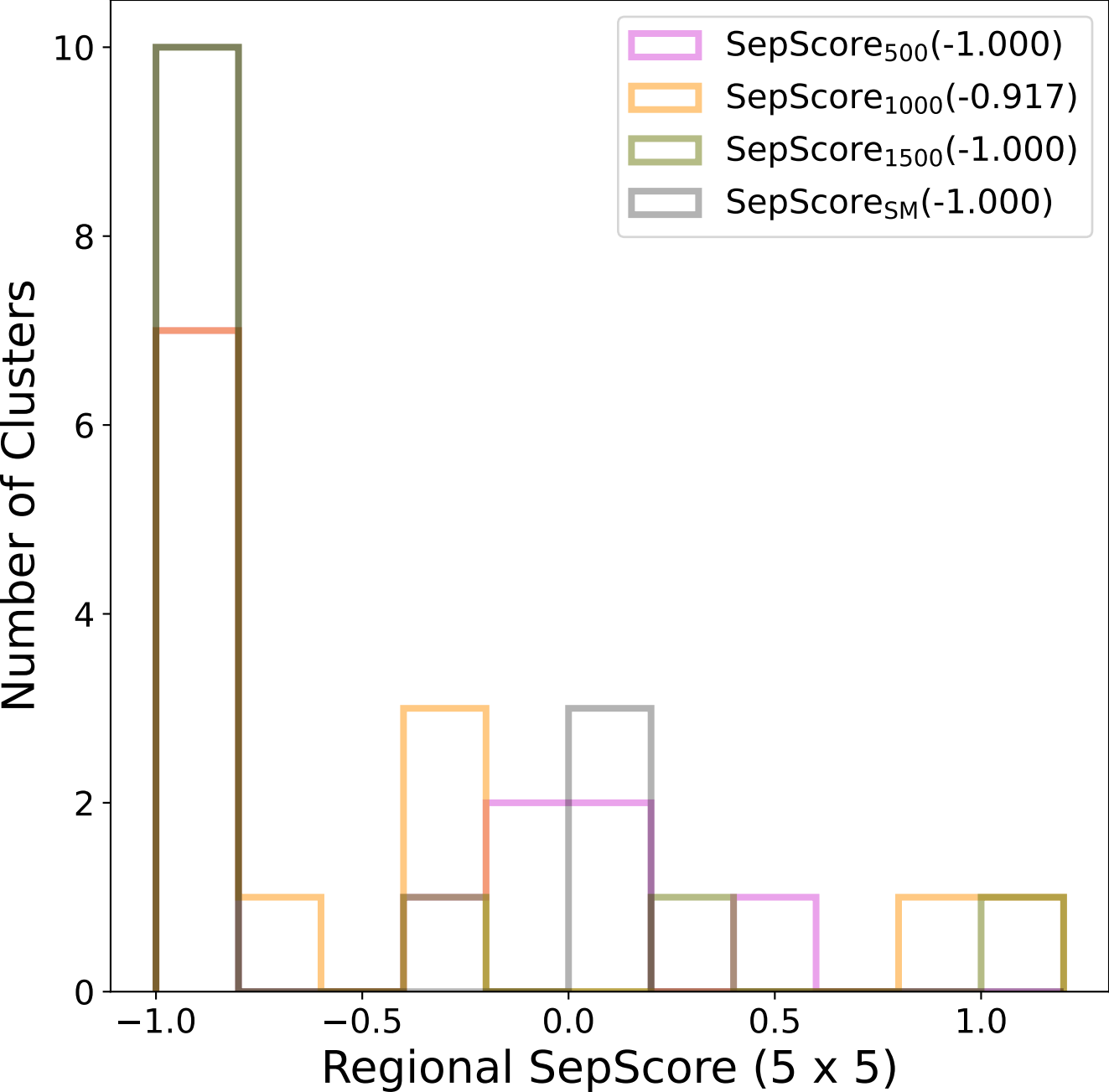
Array of label and scores BEFORE ssSM < 0.6 cut: [label, ss500, ss1000, ss1500, ssSM]:

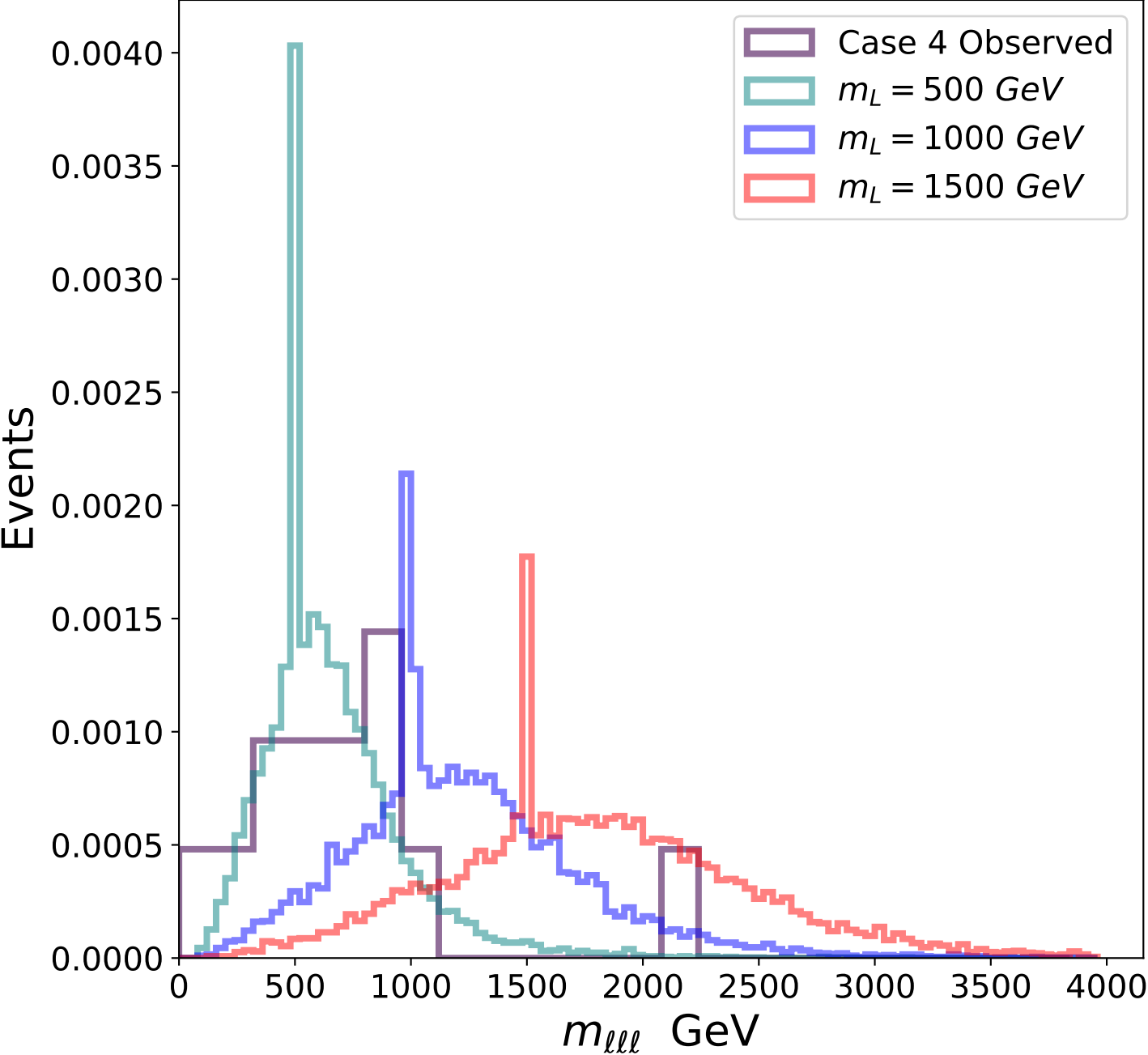
	Label	SS500	SS1000	SS1500	SSSM
0	3.0	-1.000	-1.000	-1.000	-1.000
1	4.0	-0.761	-0.999	-1.000	0.760
2	3.0	0.000	-1.000	-1.000	0.000
3	3.0	-1.000	-1.000	-1.000	-1.000
4	3.0	-1.000	-1.000	1.000	-1.000
5	4.0	-0.881	-0.999	-1.000	0.880
6	3.0	-0.333	-0.333	-0.333	-1.000
7	3.0	-1.000	-0.333	0.333	-1.000
8	4.0	-1.000	-1.000	-1.000	-1.000
9	4.0	-0.151	-0.917	-1.000	0.068
10	4.0	-0.069	-1.000	-1.000	0.069
11	3.0	-1.000	1.000	-1.000	-1.000
12	3.0	-1.000	0.926	-0.926	-1.000
13	3.0	0.500	-0.625	-1.000	-0.875
14	3.0	0.057	-0.229	-0.929	-0.900

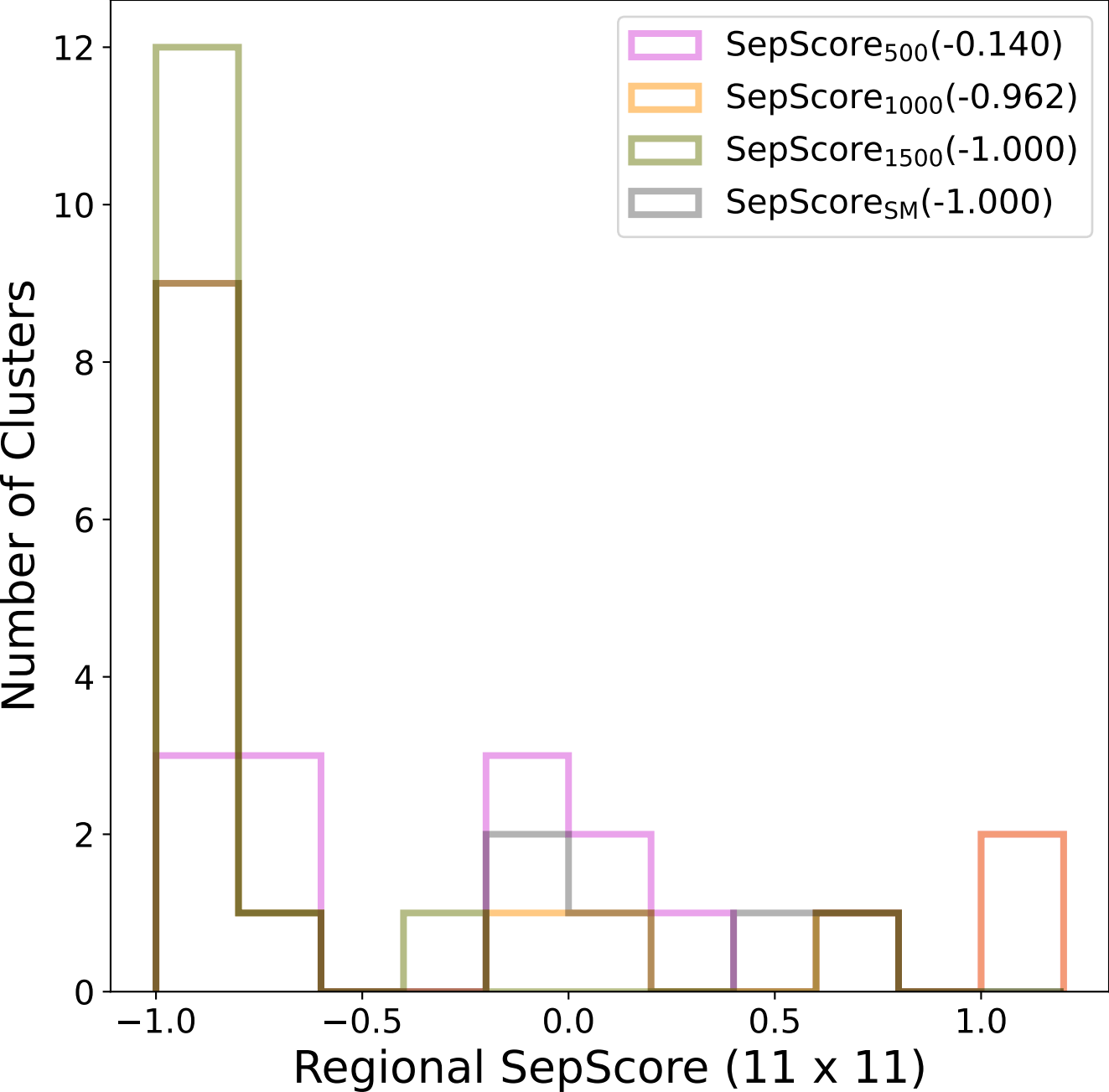
Array of label and scores AFTER ssSM < 0.6 cut: [label, ss500, ss1000, ss1500, ssSM]:

	Label	SS500	SS1000	SS1500	SSSM
0	3.0	-1.000	-1.000	-1.000	-1.000
2	3.0	0.000	-1.000	-1.000	0.000
3	3.0	-1.000	-1.000	-1.000	-1.000
4	3.0	-1.000	-1.000	1.000	-1.000
6	3.0	-0.333	-0.333	-0.333	-1.000
7	3.0	-1.000	-0.333	0.333	-1.000
8	4.0	-1.000	-1.000	-1.000	-1.000
9	4.0	-0.151	-0.917	-1.000	0.068
10	4.0	-0.069	-1.000	-1.000	0.069
11	3.0	-1.000	1.000	-1.000	-1.000
12	3.0	-1.000	0.926	-0.926	-1.000
13	3.0	0.500	-0.625	-1.000	-0.875
14	3.0	0.057	-0.229	-0.929	-0.900

Array of scores BEFORE ssSM < 0.6 cut: 15
Array of scores AFTER ssSM < 0.6 cut: 13


SepScore500(-1.000)
SepScore1000(-0.917)
SepScore1500(-1.000)
SepScoreSM(-1.000)
Number of Clusters
0
2
4
6
8
10
−1.0
−0.5
0.0
0.5
1.0
Regional SepScore (5 x 5)


Case 4 Observed
$m_L = 500\ GeV$
$m_L = 1000\ GeV$
$m_L = 1500\ GeV$
Events
0.0040
0.0035
0.0030
0.0025
0.0020
0.0015
0.0010
0.0005
0.0000
0
500
1000
1500
2000
2500
3000
3500
4000
$m_{\ell\ell\ell}$ GeV


SepScore500(-0.140)
SepScore1000(-0.962)
SepScore1500(-1.000)
SepScoreSM(-1.000)
Number of Clusters
0
2
4
6
8
10
12
−1.0
−0.5
0.0
0.5
1.0
Regional SepScore (11 x 11)

Case 4: cluster of +- 5 around the point

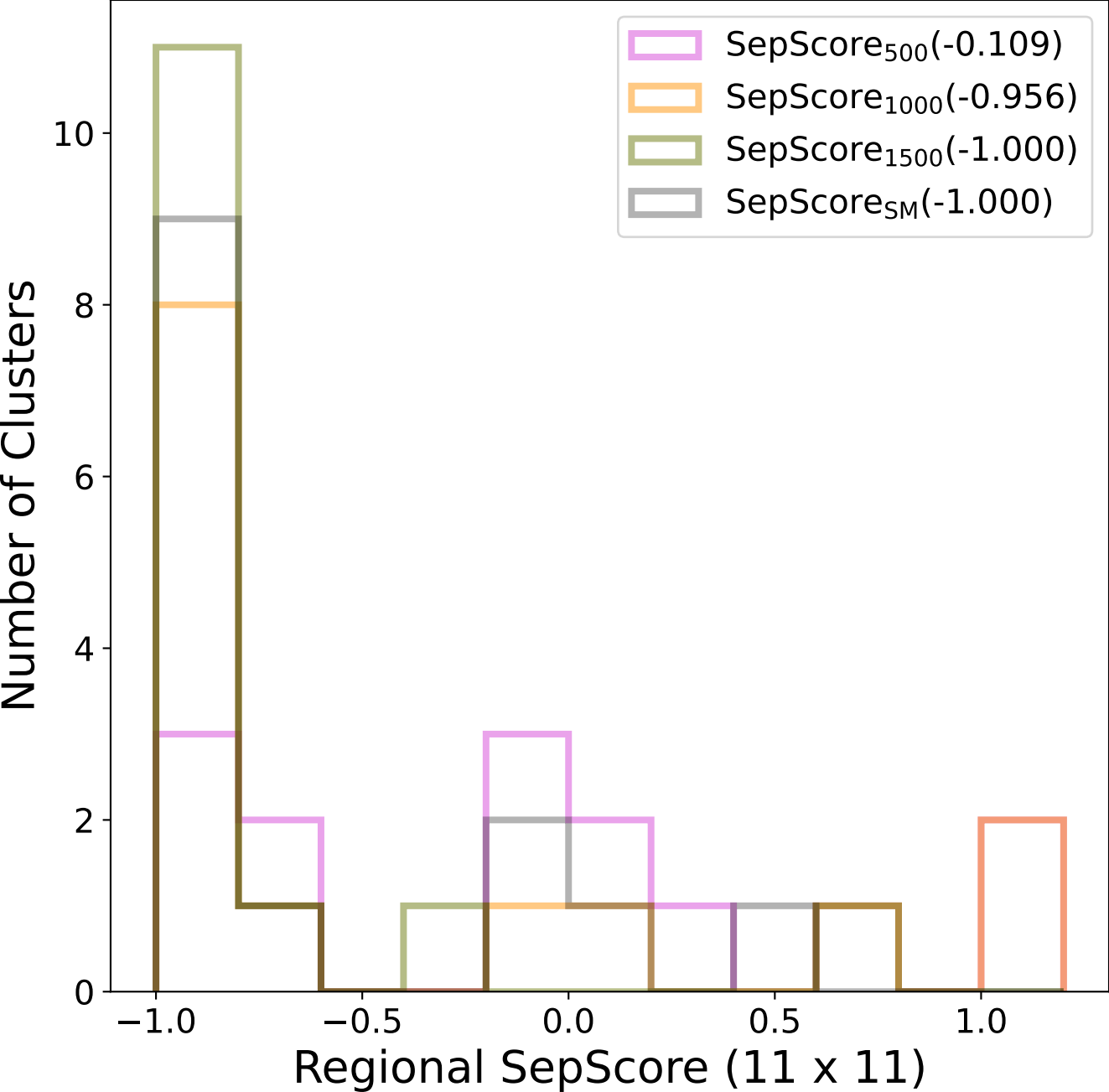
Array of label and scores BEFORE ssSM < 0.6 cut: [label, ss500, ss1000, ss1500, ssSM]:

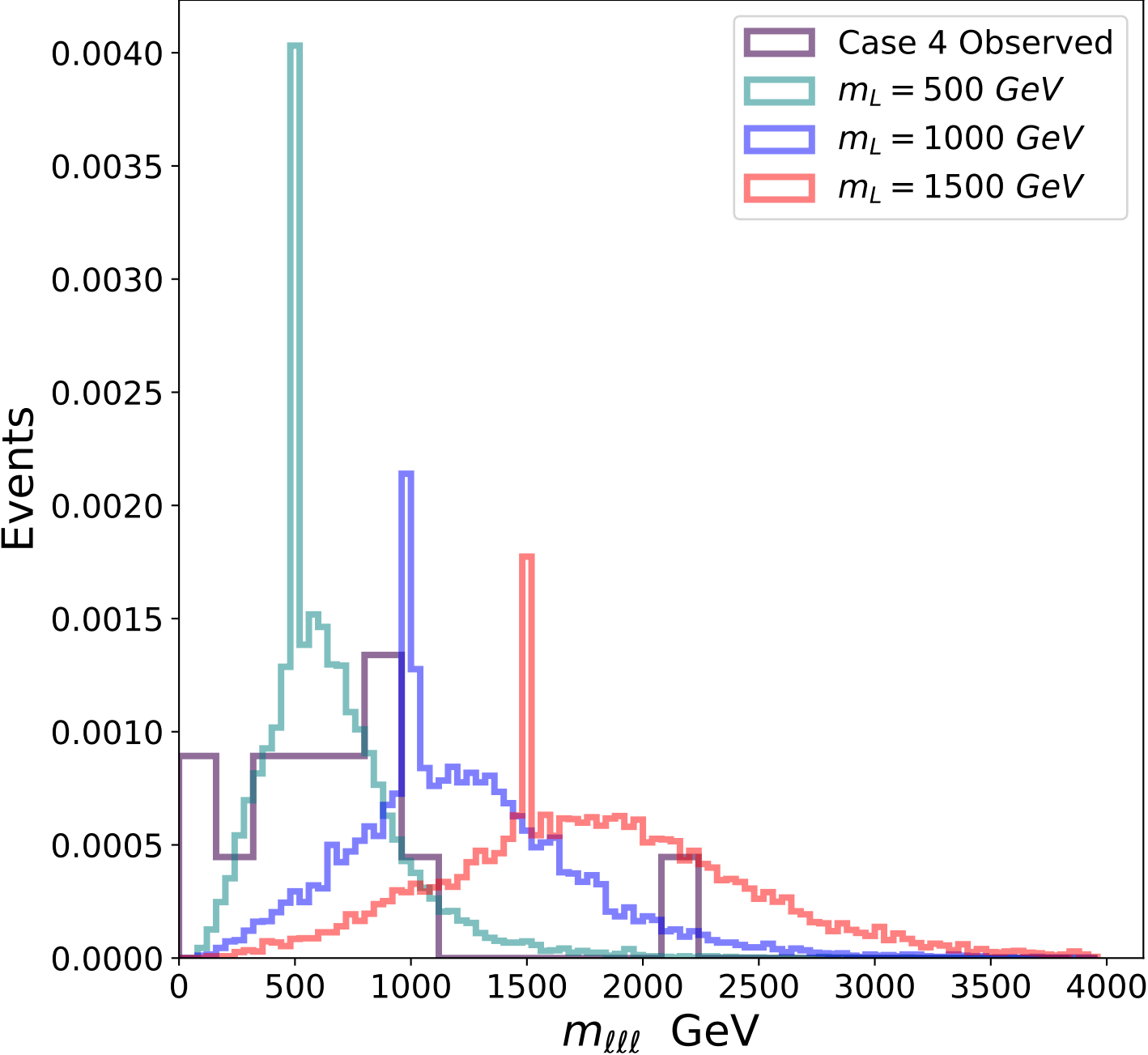
	Label	SS500	SS1000	SS1500	SSSM
0	3.0	-1.000	-0.667	0.667	-1.000
1	4.0	-0.633	-0.962	-1.000	0.595
2	3.0	0.750	-1.000	-1.000	-0.750
3	3.0	-1.000	1.000	-1.000	-1.000
4	3.0	-0.200	0.000	-0.800	-1.000
5	4.0	-0.616	-1.000	-1.000	0.616
6	3.0	0.333	-1.000	-0.333	-1.000
7	3.0	1.000	-1.000	-1.000	-1.000
8	4.0	1.000	-1.000	-1.000	-1.000
9	4.0	-0.140	-0.950	-0.988	0.078
10	4.0	0.190	-1.000	-1.000	-0.190
11	3.0	-1.000	1.000	-1.000	-1.000
12	3.0	-0.750	0.630	-0.880	-1.000
13	3.0	0.143	-1.000	-1.000	-0.143
14	3.0	-0.077	-0.191	-0.871	-0.860

Array of label and scores AFTER ssSM < 0.6 cut: [label, ss500, ss1000, ss1500, ssSM]:

	Label	SS500	SS1000	SS1500	SSSM
0	3.0	-1.000	-0.667	0.667	-1.000
1	4.0	-0.633	-0.962	-1.000	0.595
2	3.0	0.750	-1.000	-1.000	-0.750
3	3.0	-1.000	1.000	-1.000	-1.000
4	3.0	-0.200	0.000	-0.800	-1.000
6	3.0	0.333	-1.000	-0.333	-1.000
7	3.0	1.000	-1.000	-1.000	-1.000
8	4.0	1.000	-1.000	-1.000	-1.000
9	4.0	-0.140	-0.950	-0.988	0.078
10	4.0	0.190	-1.000	-1.000	-0.190
11	3.0	-1.000	1.000	-1.000	-1.000
12	3.0	-0.750	0.630	-0.880	-1.000
13	3.0	0.143	-1.000	-1.000	-0.143
14	3.0	-0.077	-0.191	-0.871	-0.860

Array of scores BEFORE ssSM < 0.6 cut: 15
Array of scores AFTER ssSM < 0.6 cut: 14


SepScore500(-0.109)
SepScore1000(-0.956)
SepScore1500(-1.000)
SepScoreSM(-1.000)
Number of Clusters
0
2
4
6
8
10
−1.0
−0.5
0.0
0.5
1.0
Regional SepScore (11 x 11)

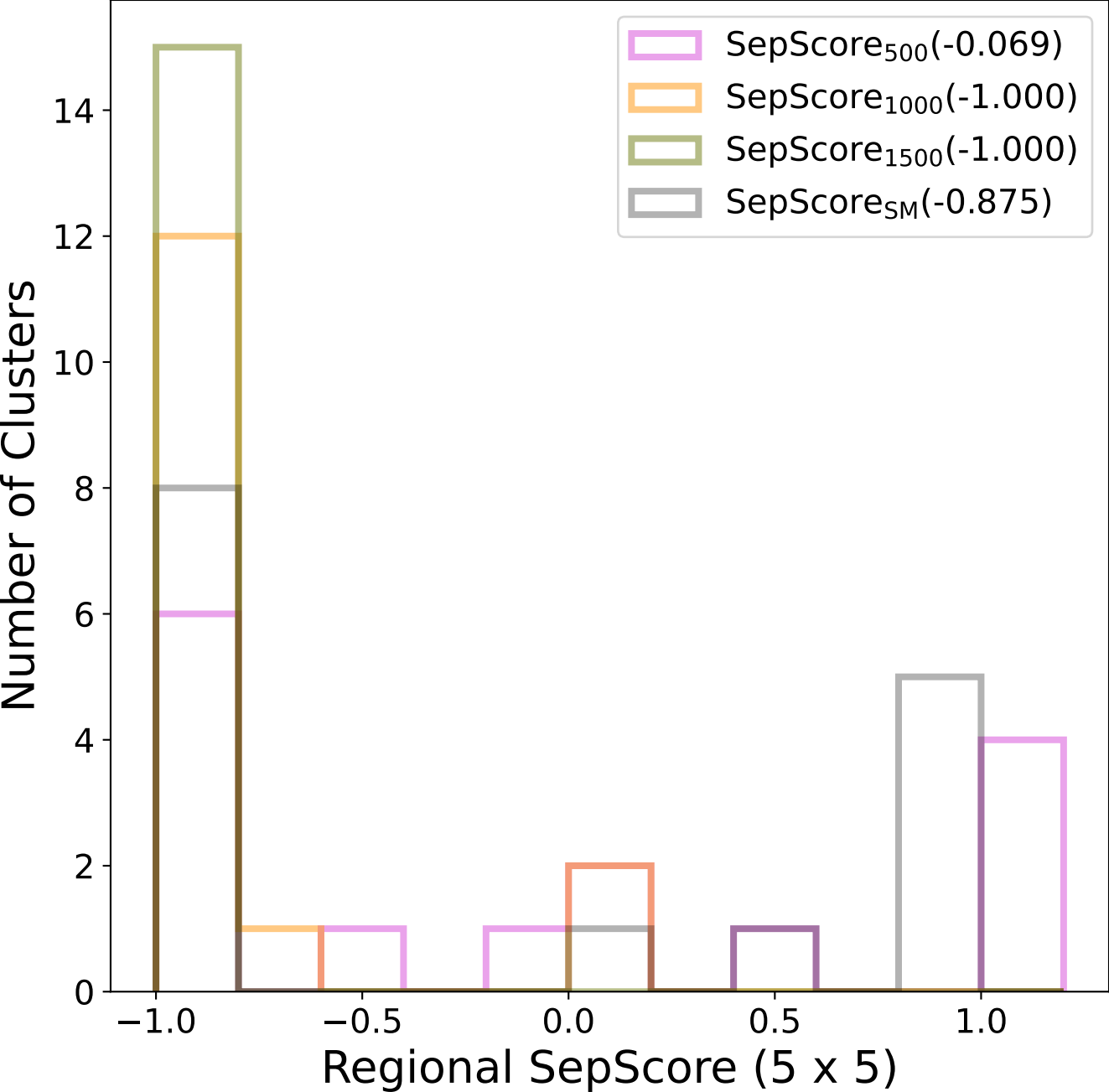

Case 4 Observed
$m_L = 500\ GeV$
$m_L = 1000\ GeV$
$m_L = 1500\ GeV$
Events
0.0040
0.0035
0.0030
0.0025
0.0020
0.0015
0.0010
0.0005
0.0000
0
500
1000
1500
2000
2500
3000
3500
4000
$m_{\ell\ell\ell}$ GeV

Case 5

Neuron Y

0 5 10 15 20 25 30 35 40 45 50 55 60 65 70 75 80 85 90 95 100

Neuron X


SepScore500(-0.069)
SepScore1000(-1.000)
SepScore1500(-1.000)
SepScoreSM(-0.875)
Number of Clusters
0
2
4
6
8
10
12
14
−1.0
−0.5
0.0
0.5
1.0
Regional SepScore (5 x 5)

Case 5: cluster of +- 2 around the point

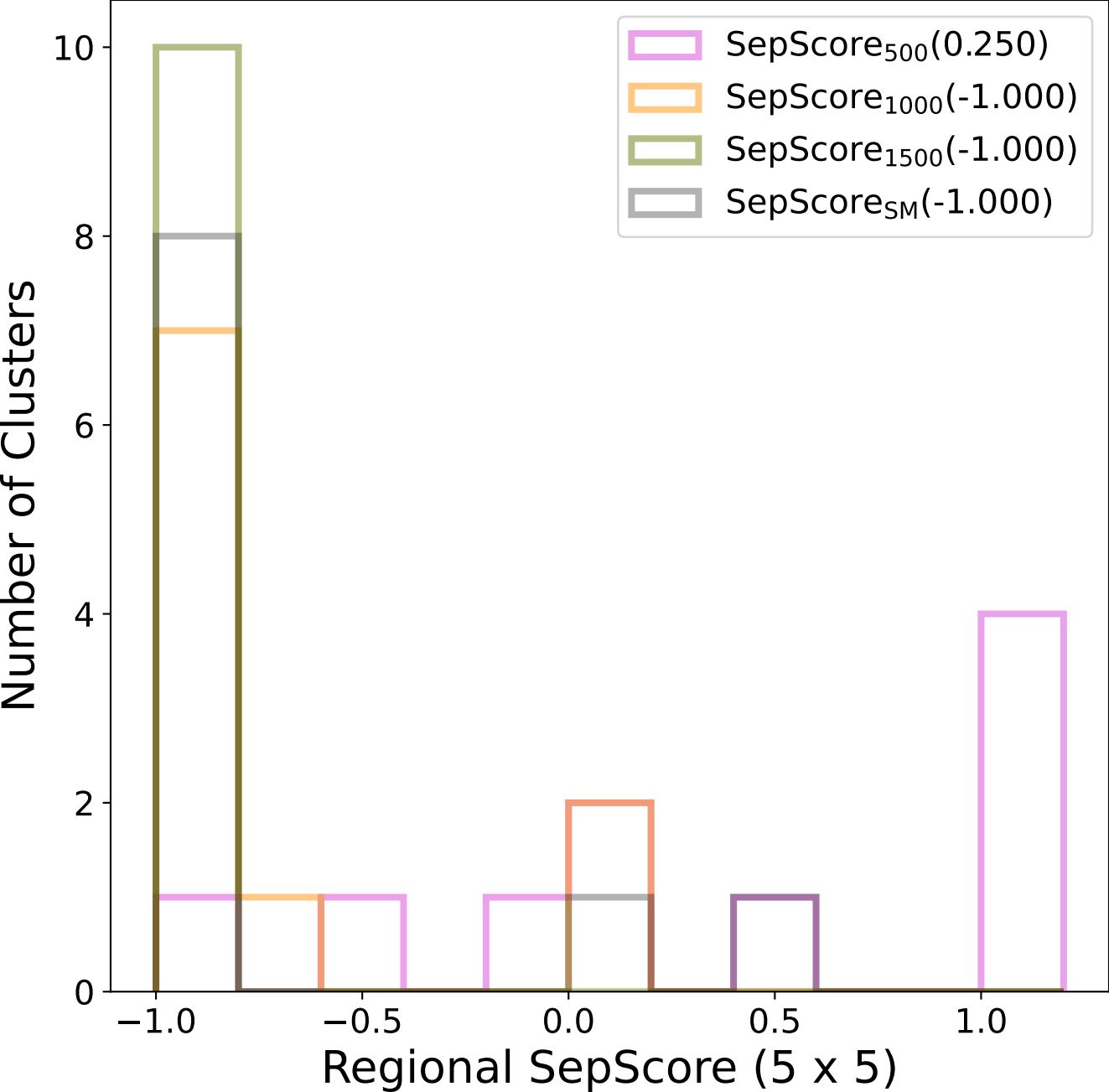
Array of label and scores BEFORE ssSM < 0.6 cut: [label, ss500, ss1000, ss1500, ssSM]:

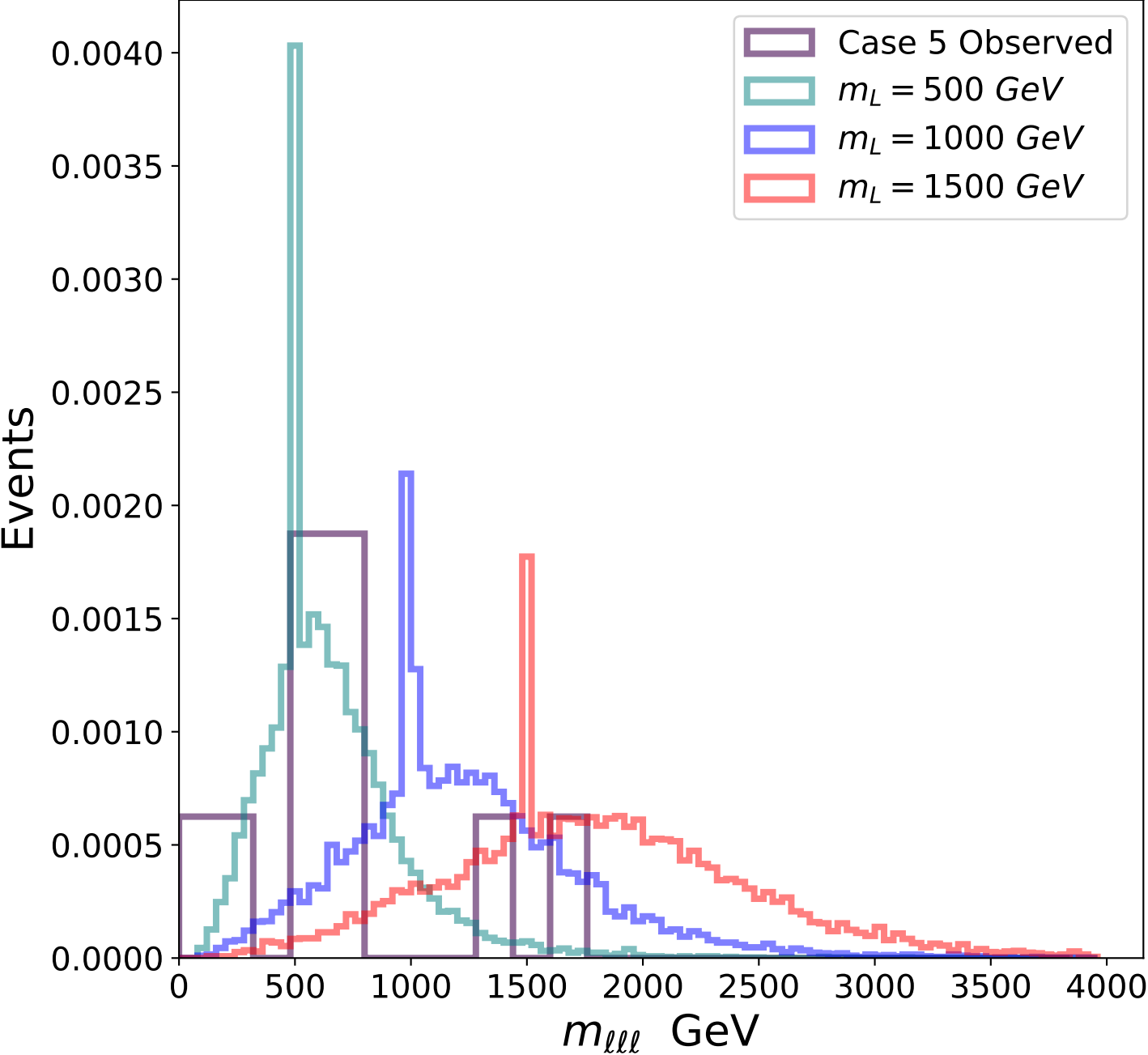
	Label	SS500	SS1000	SS1500	SSSM
0	6.0	0.000	0.000	-1.0	-1.000
1	4.0	-0.837	-0.999	-1.0	0.837
2	4.0	-0.069	-1.000	-1.0	0.069
3	6.0	1.000	-1.000	-1.0	-1.000
4	6.0	1.000	-1.000	-1.0	-1.000
5	4.0	-0.837	-0.999	-1.0	0.837
6	6.0	0.000	0.000	-1.0	-1.000
7	6.0	-0.867	-1.000	-1.0	0.867
8	6.0	0.500	-0.625	-1.0	-0.875
9	6.0	-1.000	-1.000	-1.0	-1.000
10	6.0	1.000	-1.000	-1.0	-1.000
11	4.0	-0.533	-1.000	-1.0	0.533
12	4.0	-0.932	-0.999	-1.0	0.931
13	6.0	1.000	-1.000	-1.0	-1.000
14	6.0	-0.998	-1.000	-1.0	0.998

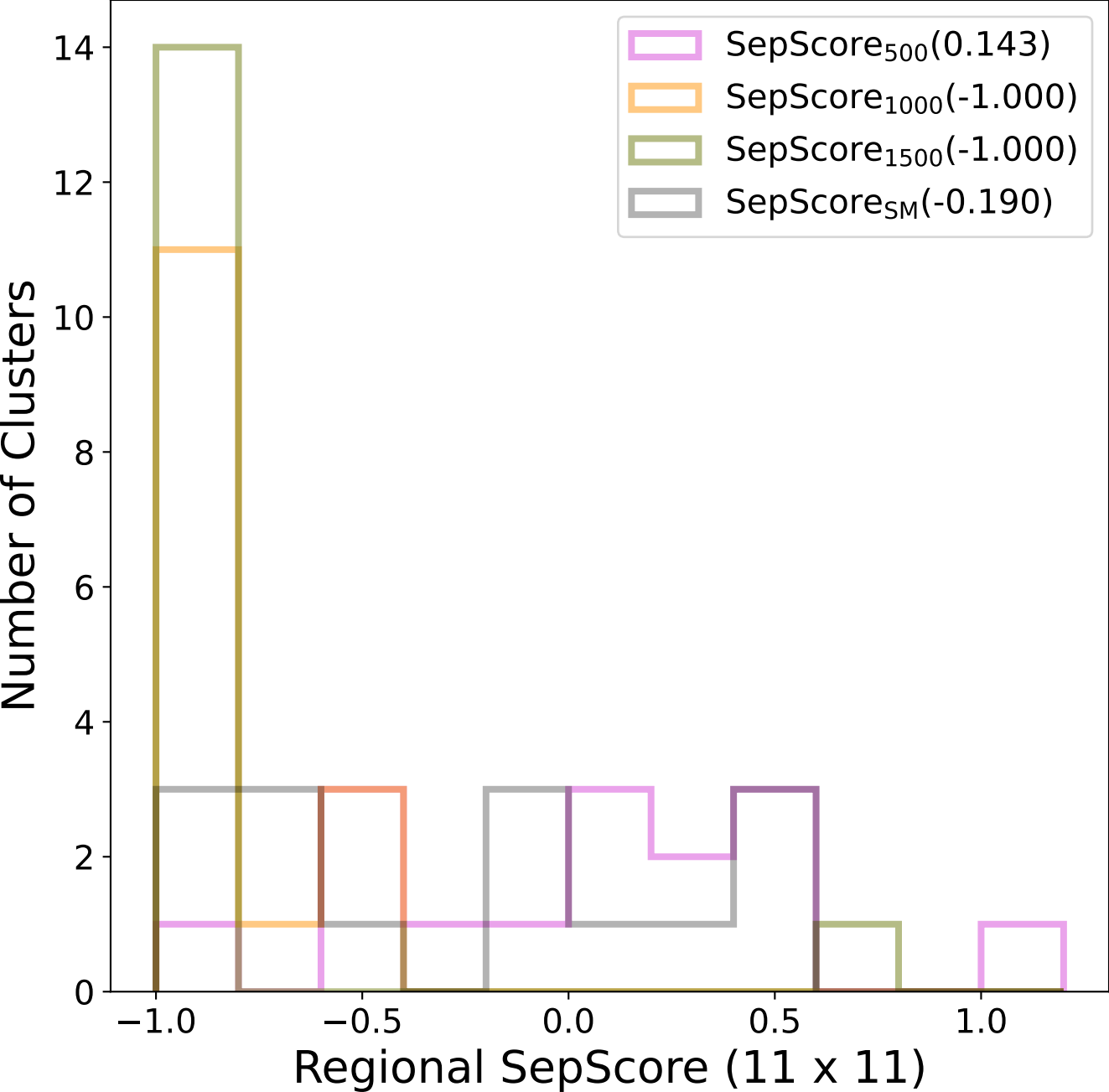
Array of label and scores AFTER ssSM < 0.6 cut: [label, ss500, ss1000, ss1500, ssSM]:

	Label	SS500	SS1000	SS1500	SSSM
0	6.0	0.000	0.000	-1.0	-1.000
2	4.0	-0.069	-1.000	-1.0	0.069
3	6.0	1.000	-1.000	-1.0	-1.000
4	6.0	1.000	-1.000	-1.0	-1.000
6	6.0	0.000	0.000	-1.0	-1.000
8	6.0	0.500	-0.625	-1.0	-0.875
9	6.0	-1.000	-1.000	-1.0	-1.000
10	6.0	1.000	-1.000	-1.0	-1.000
11	4.0	-0.533	-1.000	-1.0	0.533
13	6.0	1.000	-1.000	-1.0	-1.000

Array of scores BEFORE ssSM < 0.6 cut: 15
Array of scores AFTER ssSM < 0.6 cut: 10


SepScore500(0.250)
SepScore1000(-1.000)
SepScore1500(-1.000)
SepScoreSM(-1.000)
Number of Clusters
0
2
4
6
8
10
−1.0
−0.5
0.0
0.5
1.0
Regional SepScore (5 x 5)


Case 5 Observed
$m_L = 500\ GeV$
$m_L = 1000\ GeV$
$m_L = 1500\ GeV$
Events
0.0040
0.0035
0.0030
0.0025
0.0020
0.0015
0.0010
0.0005
0.0000
0
500
1000
1500
2000
2500
3000
3500
4000
$m_{\ell\ell\ell}$ GeV


SepScore500(0.143)
SepScore1000(-1.000)
SepScore1500(-1.000)
SepScoreSM(-0.190)
Number of Clusters
0
2
4
6
8
10
12
14
−1.0
−0.5
0.0
0.5
1.0
Regional SepScore (11 x 11)

Case 5: cluster of +- 5 around the point

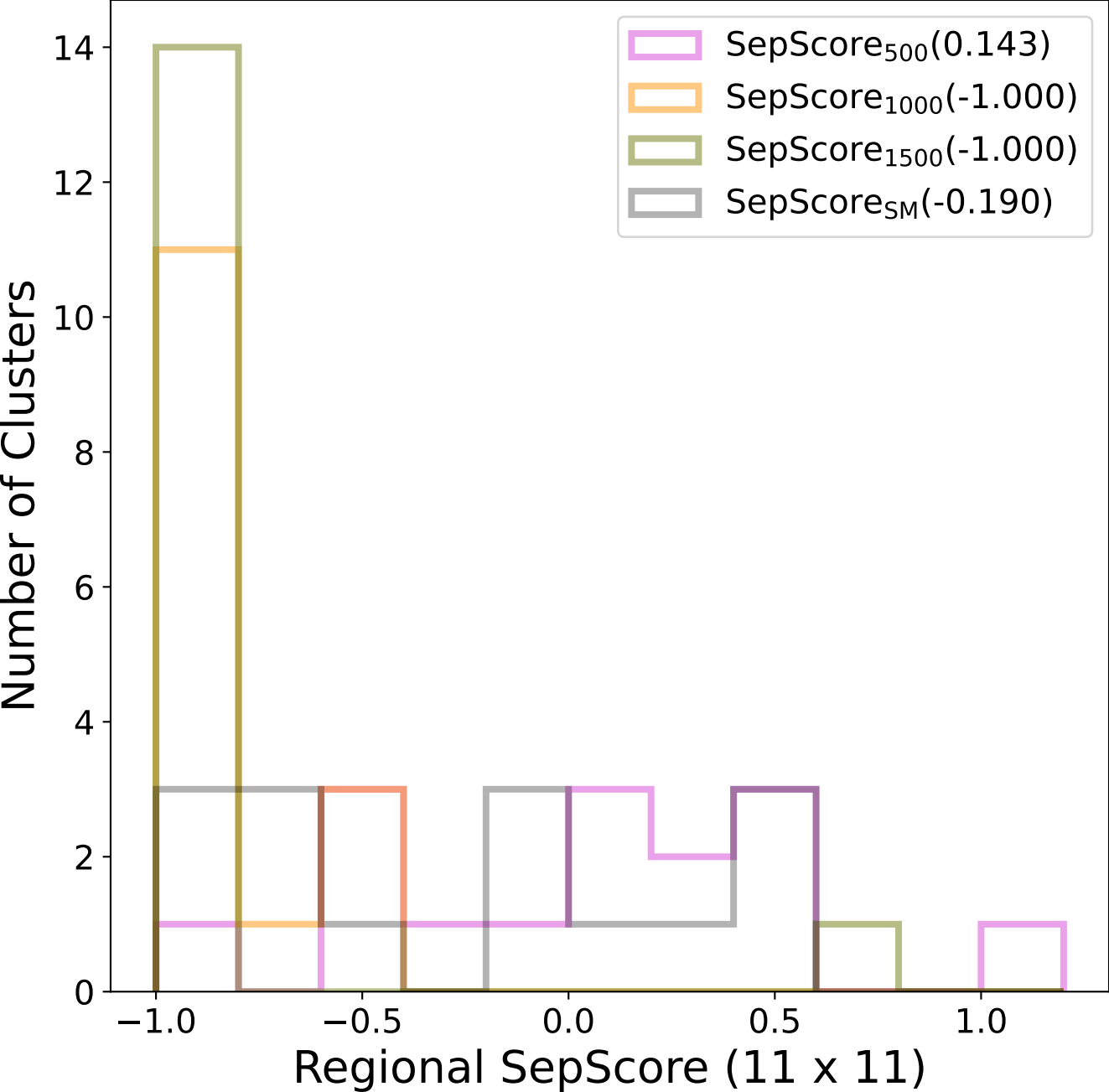
Array of label and scores BEFORE ssSM < 0.6 cut: [label, ss500, ss1000, ss1500, ssSM]:

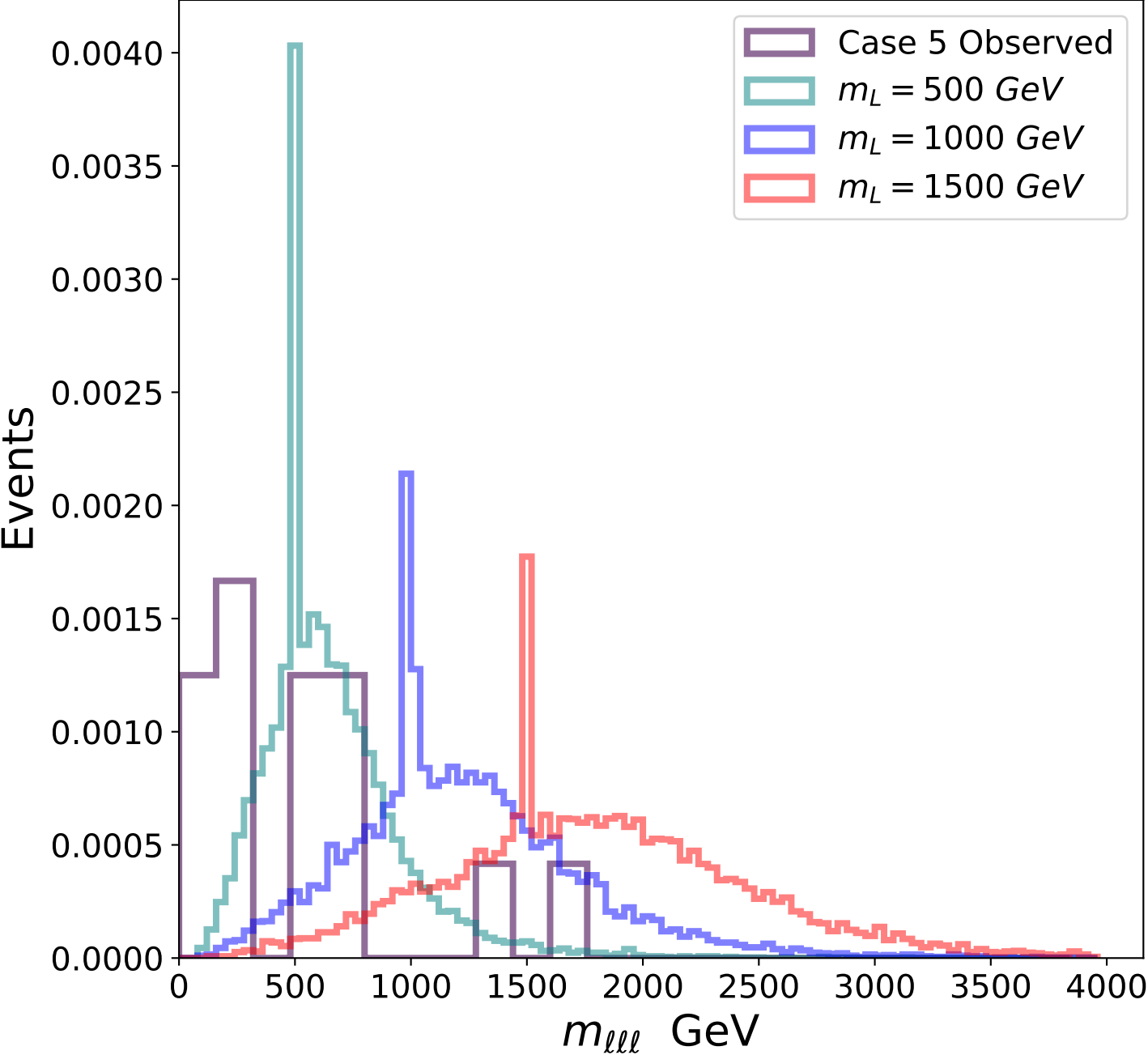
	Label	SS500	SS1000	SS1500	SSSM
0	6.0	0.259	-0.481	-1.000	-0.778
1	4.0	-0.571	-0.977	-1.000	0.548
2	4.0	0.190	-1.000	-1.000	-0.190
3	6.0	0.540	-0.850	-0.980	-0.710
4	6.0	0.556	-1.000	-1.000	-0.556
5	4.0	-0.571	-0.977	-1.000	0.548
6	6.0	0.500	-0.500	-1.000	-1.000
7	6.0	0.028	-1.000	-1.000	-0.029
8	6.0	0.143	-1.000	-1.000	-0.143
9	6.0	-1.000	-0.667	0.667	-1.000
10	6.0	0.222	-0.556	-1.000	-0.667
11	4.0	-0.148	-1.000	-1.000	0.148
12	4.0	-0.495	-1.000	-1.000	0.495
13	6.0	1.000	-1.000	-1.000	-1.000
14	6.0	-0.306	-1.000	-1.000	0.306

Array of label and scores AFTER ssSM < 0.6 cut: [label, ss500, ss1000, ss1500, ssSM]:

	Label	SS500	SS1000	SS1500	SSSM
0	6.0	0.259	-0.481	-1.000	-0.778
1	4.0	-0.571	-0.977	-1.000	0.548
2	4.0	0.190	-1.000	-1.000	-0.190
3	6.0	0.540	-0.850	-0.980	-0.710
4	6.0	0.556	-1.000	-1.000	-0.556
5	4.0	-0.571	-0.977	-1.000	0.548
6	6.0	0.500	-0.500	-1.000	-1.000
7	6.0	0.028	-1.000	-1.000	-0.029
8	6.0	0.143	-1.000	-1.000	-0.143
9	6.0	-1.000	-0.667	0.667	-1.000
10	6.0	0.222	-0.556	-1.000	-0.667
11	4.0	-0.148	-1.000	-1.000	0.148
12	4.0	-0.495	-1.000	-1.000	0.495
13	6.0	1.000	-1.000	-1.000	-1.000
14	6.0	-0.306	-1.000	-1.000	0.306

Array of scores BEFORE ssSM < 0.6 cut: 15
Array of scores AFTER ssSM < 0.6 cut: 15


SepScore500(0.143)
SepScore1000(-1.000)
SepScore1500(-1.000)
SepScoreSM(-0.190)
Number of Clusters
0
2
4
6
8
10
12
14
−1.0
−0.5
0.0
0.5
1.0
Regional SepScore (11 x 11)


Case 5 Observed
$m_L = 500\ GeV$
$m_L = 1000\ GeV$
$m_L = 1500\ GeV$
Events
0.0040
0.0035
0.0030
0.0025
0.0020
0.0015
0.0010
0.0005
0.0000
0
500
1000
1500
2000
2500
3000
3500
4000
$m_{\ell\ell\ell}$ GeV